Contents

1 Why we estimate this equation: the HSA Phillips curve

Our estimating equation is the theory-motivated counterpart of the New Keynesian Phillips curve derived by Fujiwara–Matsuyama (2026, *Journal of Monetary Economics*) under the HSA demand system. This section states their result, maps it to what we take to data, and makes explicit how far the empirical specification departs from the theory.

1.1 The HSA Phillips curve

A final-good producer aggregates a continuum of varieties through an HSA demand system with market-share function $s(z)$, where $z = p/A$ is a firm's price normalized by the single price aggregator A (the reference price faced by firms, an inverse index of competitive pressure). Intermediate firms set prices under a Rotemberg adjustment cost χ and enter endogenously with a one-period time-to-build lag. Solving the firm's problem, PPI inflation follows (their Eq. 21)

$$\hat{\pi}_t = \beta(1-\delta)\,\mathbb{E}_t\hat{\pi}_{t+1} + \underbrace{\frac{\zeta(z)-1}{\chi}}_{\text{slope on marginal cost}} \left(\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t\right) - \underbrace{\frac{1}{\chi}\frac{1-\rho(z)}{\rho(z)}}_{\text{entry / cost-push}} \hat{N}_t, \tag{1}$$

where $\hat{N}_t$ is the inverse HHI (number of firms), $\hat{W}_t - \hat{Z}_{P,t} - \hat{p}_t$ is the inverse markup (the model's measure of real marginal cost), $\zeta(z) = 1 - s'(z)z/s(z) > 1$ is the price-elasticity function (so the flexible-price markup is $\mu = \zeta/(\zeta-1)$), and $\rho(z)$ is the pass-through rate.

1.2 Two ways competition enters

(1) The slope is a function of the firm count (structural flattening). The coefficient on marginal cost, $(\zeta(z)-1)/\chi$, depends on the price elasticity $\zeta(z)$, which through $z = p/A$ and the symmetric-equilibrium condition $s(z) = 1/N$ is a function of the firm count N. Rising concentration—*fewer firms*, higher markups, lower pass-through $\rho(z)$—lowers this slope, so the Phillips curve flattens *structurally*. Linearizing the slope in the trend level N, $\bar{N}_t$ of the firm count gives

$$\kappa_t = \kappa_0 + \delta\,\bar{N}_t, \tag{2}$$

so a *positive* δ is the theory's flattening prediction: the slope moves with the firm count and falls as concentration rises and N declines.

(2) Endogenous entry is a cost-push term (estimation bias). The second term $-\frac{1}{\chi}\frac{1-\rho(z)}{\rho(z)}\hat{N}_t$ appears only away from CES. It means that the cyclical component of firm entry acts as a cost-push shock through strategic complementarity. Omitting it biases an empirical Phillips-curve estimate (omitted-variable problem). We retain the cyclical term explicitly with its own coefficient $\theta_t = \theta_0 + \gamma\,\bar{N}_t$.

CES is the irrelevant case. Under CES, $s(z) = \gamma_{\text{CES}}\,z^{1-\theta}$, so $\zeta(z) = \theta$ is constant and pass-through is $\rho(z) = 1$. The entry term vanishes and the slope is the constant $(\theta-1)/\chi$: market concentration neither flattens the curve nor moves inflation. This is exactly our CES benchmark, and departing from CES (the HSA models) is what can make δ and the $\hat{N}_t$ term nonzero.

1.3 From theory to the estimating equation

Writing the activity/inverse-markup gap as x_t, adding a backward-looking inertia term (α) and a reduced-form shock e_t, and decomposing the observed firm count into trend and cycle,

$$N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t, \tag{3}$$

where $\bar{N}_t$ is a random-walk-with-drift trend, $\hat{N}_t$ is a stationary AR(2) gap (truncated to the stationary region), and $\nu_t \sim \mathcal{N}(0, \sigma_N^2)$ is measurement error, the HSA Phillips curve becomes the equation we estimate:

$$\boxed{\pi_t = \alpha\, \pi_{t-1} + (1-\alpha)\, \mathbb{E}_t \pi_{t+1} + \kappa_t\, x_t - \theta_t\, \hat{N}_t + e_t, \qquad \kappa_t = \kappa_0 + \delta\, \bar{N}_t, \;\; \theta_t = \theta_0 + \gamma\, \bar{N}_t.} \tag{4}$$

The trend $\bar{N}_t$ carries the structural-flattening channel (through δ); the cycle $\hat{N}_t$ carries the entry cost-push channel (through θ_t).

1.4 Distance from the theory

We call this exercise *semi-structural* rather than structural because the specification departs from (1) in several ways. To avoid over-reading the coefficients, we state each:

- the theory is about *PPI* inflation, but there is no PPI expectation series, so we pair observed PPI with a common external expectation series and the expectation term is not a matched structural object;
- the theoretical regressor is real marginal cost / the inverse markup, but the main results use the negative unemployment gap.
- apart from the hybrid-NKPC normalization that puts weights α and $1-\alpha$ on lagged and expected inflation, we impose no structural cross-coefficient restrictions from the HSA theory and approximate κ_t as a *linear* function of $\bar{N}_t$;
- we add backward-looking inertia (α), which is absent from (1);
- expected inflation is an external survey measure (Section 2);

2 Data

Sample. Quarterly, 1982Q1–2012Q4 ($T = 124$) in the US. These labels and the observation count are read from the selected run's metadata when the report is rebuilt; they are not maintained independently in the prose.

Inflation. We compare headline CPI, core CPI, and PPI. Expected inflation $\mathbb{E}\pi$ is the same external survey series in all specifications. Because the Survey of Professional Forecasters does not publish a PPI forecast, the PPI specifications use the same expectation series as CPI. The PPI estimates are therefore not a structural estimate matched through the expectation term, but a semi-structural external-validity check that changes only the observed inflation index. A measurement mismatch with the expectation series also remains for core CPI.

Headline CPI and PPI are simple four-quarter percentage changes; core CPI is a four-quarter log difference. All three are sampled quarterly, so adjacent observations overlap in three of four quarters. Consequently the backward-looking coefficient α mixes genuine inertia with mechanical overlap, while the production likelihood treats the remaining inflation disturbance as serially independent. The companion `error_robustness` implementation replaces it with an invertible MA(3), the order implied by this overlap. Its full production sweep is deliberately not folded into the headline tables until those runs have been completed; the current headline posterior is therefore the i.i.d.-disturbance specification throughout, not a mixture of the two.

Activity gaps (five). Negative unemployment gap, BN output gap, HP output gap (100× log real output HP-filtered, in the same 100-log-point units as BN), HP labor-share gap, and inverse markup. The unemployment measure enters as the *negative* unemployment gap $x_t = -(u_t - u_t^*)$—labor-market tightness, positive in expansions and negative in downturns (e.g. $x_t \approx -4$ in 2009), co-moving positively with the output gaps. A positive slope κ on x_t is therefore a conventional downward-sloping Phillips curve.

Competition proxy, and why the main results are mixed-frequency. The competition series is Gustavo's firm-count measure, i.e. the inverse of the *HHI of U.S. listed firms*. It is **annual**: over 1982–2012 there are 31 observations, not 124.

The main results therefore use a *mixed-frequency* observation scheme. The annual value enters the state-space model in that year's Q4 and the intervening 93 quarters are treated as *missing*, so the sub-annual firm count is inferred rather than assumed (Section 3.3). Alongside it we report the conventional alternative, in which the annual series is PCHIP-interpolated to a quarterly value and that interpolated value is then used *as data* in all 124 quarters (Section 7).

Earlier revisions of this paper took the interpolated version as the main result. We reversed that after finding that the interpolation is not innocuous. It manufactures within-year variation that does not exist, drives the estimated firm-count measurement error down to $\sigma_N \approx 0.023$ (versus 0.069 under mixed frequency), and through that pins $\bar{N}_t + \hat{N}_t$ so tightly at all 124 quarters that only the *split* between trend and cycle is left free. The consequence is a near-exact posterior ridge between the two states—posterior correlation -0.9996 under interpolation against $+0.1957$ under mixed frequency—which is the identification pathology described in Section 8. The interpolated design also forces the AR(2) cycle to a near-unit root and hence to a 94.0-quarter "cycle" that is a second trend rather than a cycle. None of this is visible if the interpolated series is simply taken as data.

The reassuring part is that δ itself barely moves between the two designs, so the substantive result does not hinge on the choice; what changes is how much of the reported precision is real.

Unit convention. The default transform is $(100 \log N - \overline{100 \log N})/10$, so one unit of $\bar{N}$ or $\hat{N}$ is a ten-log-point change around the sample mean. The reported $\delta, \theta, \theta_0, \gamma$ are all on this ten-log-point scale.

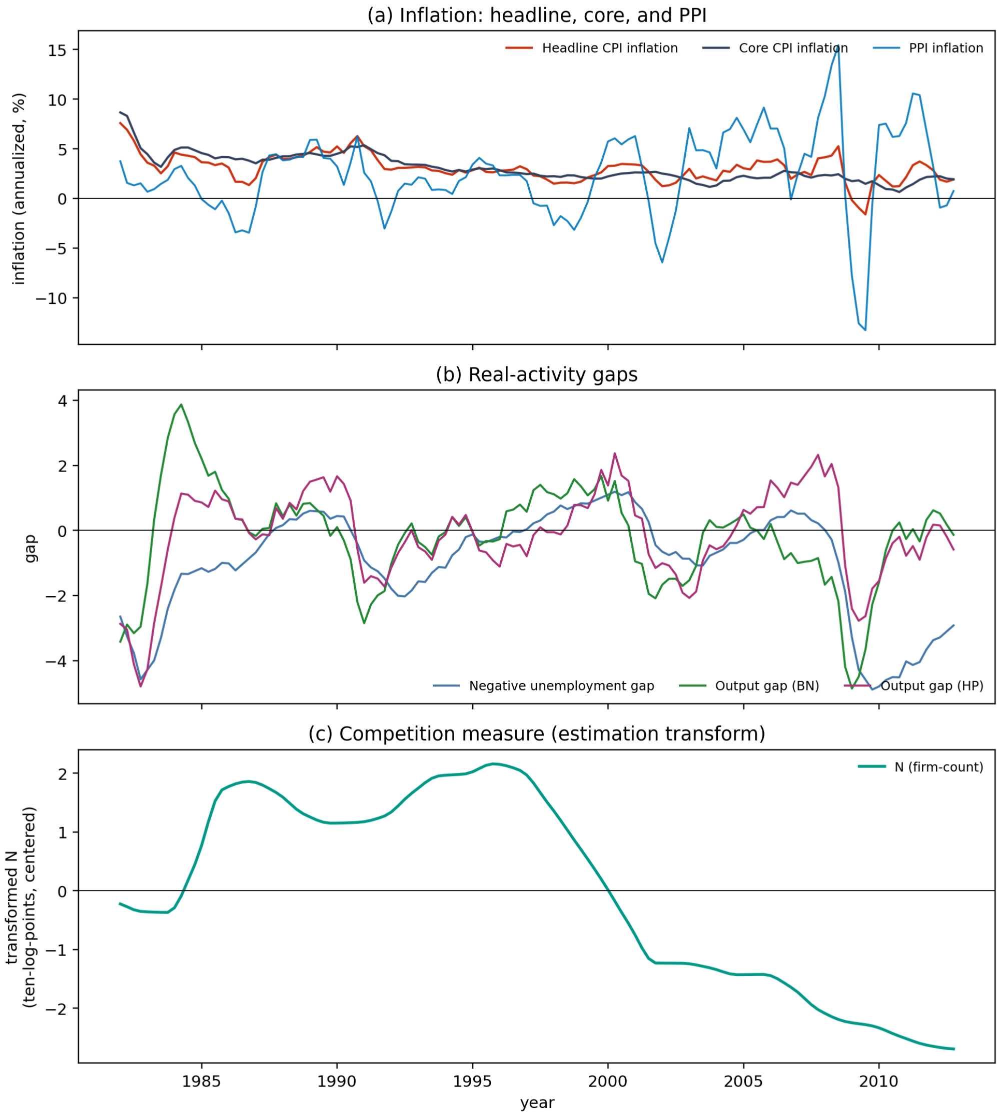

Figure 1: Data. (a) Headline, core, and PPI inflation. (b) Real-activity gaps (negative unemployment gap, BN and HP output gaps). (c) Competition measure (estimation transform of the listed-firm count).

3 Econometric model and estimation

3.1 Model variants

The five models differ only in the restriction on the slope (Table 1).

Table 1: Model variants.

Model	κ_t	θ_t	Role	State block
CES	constant	0	Textbook NKPC benchmark	Gibbs (conjugate)
HSA steady	$\kappa_0 + \delta\bar{N}_t$	0	$\bar{N}$ channel only	Gibbs + exact joint FFBS
HSA dynamic	constant	constant	$\hat{N}$ channel only	Gibbs + exact joint FFBS
HSA const-theta	$\kappa_0 + \delta\bar{N}_t$	constant	HSA full with $\gamma = 0$	Gibbs + exact joint FFBS
HSA full	$\kappa_0 + \delta\bar{N}_t$	$\theta_0 + \gamma\bar{N}_t$	HSA full	Gibbs + Particle Gibbs

State blocks. The bilinear term $-\gamma\bar{N}_t\hat{N}_t$ makes the joint firm-count state non-linear-Gaussian, so **HSA full is estimated by a Particle Gibbs sampler**: one conditional sequential-Monte-Carlo sweep updates the whole state path (512 particles). This applies to *every* HSA-full cell, under both the PCHIP and the annual-Q4 observation scheme, so the PCHIP-versus-annual-Q4 comparison of Section 7 differs only in the observation scheme and not in the sampler. The table builder refuses to emit a table in which any HSA-full cell was drawn by a different sampler.

Every other model draws the whole state $s_t = (\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)'$ jointly by exact Kalman/FFBS. That includes **HSA const-theta**: with $\gamma = 0$ the inflation observation is linear in the joint state, so the model is linear-Gaussian and admits an exact joint draw. Earlier revisions sampled it with the same two alternating blocks as HSA full; because $\bar{N}_t$ and $\hat{N}_t$ are almost perfectly negatively correlated in the posterior under interpolation (-0.9996: the firm-count equation pins their sum but barely their split), that blocking moved their common level with an integrated autocorrelation time of order 10^3 sweeps and did not mix. This is a purely computational change—it removes a sampler pathology and creates no identification that the data do not contain. Coefficients, variances, priors, chains, iterations, burn-in and thinning are unchanged throughout.

3.2 The inflation equation actually estimated

To put the theoretical equation (4) directly in regression form, we move expected inflation to the left-hand side and define

$$y_t \equiv \pi_t - E_t\pi_{t+1}, \qquad a_t \equiv \pi_{t-1} - E_t\pi_{t+1} \tag{5}$$

Making the unforecastable component of the activity variable explicit, the estimating system is

$$y_t = \alpha a_t + (\kappa_0 + \delta\bar{N}_t)x_t - (\theta_0 + \gamma\bar{N}_t)\hat{N}_t + \lambda_{e\zeta}\zeta_t + \eta_t, \tag{6}$$

$$x_t = \phi_1 x_{t-1} + \zeta_t, \qquad \eta_t \perp \zeta_t. \tag{7}$$

That is, the original NKPC shock decomposes as $e_t = \lambda_{e\zeta}\zeta_t + \eta_t$. Estimating $\lambda_{e\zeta}$ allows a correlation between the contemporaneous activity shock and the inflation shock, rather than forcing a simple regression that treats x_t as exogenous. The saved correlation coefficient is

$$\rho_{e\zeta} = \frac{\lambda_{e\zeta}\sigma_\zeta}{\sqrt{\lambda_{e\zeta}^2\sigma_\zeta^2 + \sigma_\eta^2}} \tag{8}$$

The main HSA steady sets $\theta_0 = \gamma = 0$; HSA dynamic sets $\delta = \gamma = 0$.

For numerical stability the implementation updates internal coefficients on the rescaled regressors $x_t/100$ and $x_t\bar{N}_t/100$ for κ_0 and δ; the coefficients are rescaled back by 100 on saving, so the $\kappa_0, \delta, \kappa_t$ in the text, tables, and figures are all in the physical units of (6). They must not be multiplied by 100 or divided by 100 again.

3.3 State-space representation splitting the firm count into trend and cycle

The observation and transition equations for the firm count are

$$N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t, \qquad \nu_t \sim N(0, \sigma_N^2), \tag{9}$$

$$\hat{N}_t = \rho_1\hat{N}_{t-1} + \rho_2\hat{N}_{t-2} + u_t, \qquad u_t \sim N(0, \sigma_u^2), \tag{10}$$

$$\bar{N}_t = n + \bar{N}_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma_\varepsilon^2) \tag{11}$$

where $\bar{N}_t$ is the low-frequency trend with drift n and $\hat{N}_t$ is a stationary AR(2) cycle. Rather than a two-step procedure that first filters the observed firm count into components and then substitutes them into the NKPC, we extract both states using the firm-count equation and the inflation equation within the same Gibbs iteration.

Two observation frequencies; the main results use the first. The firm count is annual, so the *mixed-frequency* design places the annual value in each year's Q4 and treats Q1–Q3 as missing,

$$N_t^{\text{obs}} = \begin{cases} (100\log N_{y(t)}^{\text{annual}} - \bar{c})/10, & t \in \mathcal{T}_N \text{ (Q4)}, \\ \text{missing}, & t \notin \mathcal{T}_N \text{ (Q1--Q3)}, \end{cases} \qquad |\mathcal{T}_N| = 31, \tag{12}$$

where $\bar{c}$ is a fixed centering constant, shared with the interpolated series so the coefficient units are comparable across designs. The alternative *interpolated* design instead PCHIP-interpolates the annual series to quarterly and applies (9) at all 124 quarters; Section 7 reports it and documents what the interpolation costs. In Q4 the Kalman update uses two observation rows—the firm-count equation and the inflation equation—and in Q1–Q3 only the one inflation row. Because the state transition and prediction continue at every quarter, the Q1–Q3 $\bar{N}_t, \hat{N}_t$ are not simple interpolations but FFBS draws conditioned on the annual anchors, the state transition, and the inflation equation, updated every iteration. The inverse-gamma update of σ_N^2 likewise uses only the 31 finite Q4 residuals.

For the main HSA steady the state is

$$s_t = (\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)', \qquad s_t = c + Fs_{t-1} + w_t \tag{13}$$

with

$$c = \begin{bmatrix} 0 \\ 0 \\ n \end{bmatrix}, \quad F = \begin{bmatrix} \rho_1 & \rho_2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad Q = \mathrm{Var}(w_t) = \begin{bmatrix} \sigma_u^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma_\varepsilon^2 \end{bmatrix}. \tag{14}$$

Stacking the firm-count observation with the inflation equation net of its known part,

$$y_t^\pi = y_t - \alpha a_t - \kappa_0 x_t - \lambda_{e\zeta}\zeta_t \tag{15}$$

gives

$$\underbrace{\begin{bmatrix} N_t^{\mathrm{obs}} \\ y_t^\pi \end{bmatrix}}_{z_t} = \underbrace{\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & \delta x_t \end{bmatrix}}_{H_t} s_t + v_t, \qquad \mathrm{Var}(v_t) = \begin{bmatrix} \sigma_N^2 & 0 \\ 0 & \sigma_\eta^2 \end{bmatrix}. \tag{16}$$

Conditional on the coefficients and variances this is a linear-Gaussian state-space model, so $s_{0:T}$ can be drawn jointly by an exact Kalman filter / forward-filtering backward-sampling (FFBS).

The initial state starts from the fixed prior

$$s_0 = (\hat{N}_0, \hat{N}_{-1}, \bar{N}_0)' \sim N(m_0, P_0), \qquad m_0 = (0,0,0)', \quad P_0 = 10 I_3 \tag{17}$$

We do not substitute N_0^{obs} into m_0; the period-0 firm count and inflation enter the ordinary Kalman observation update. The first AR(2) transition also actually uses the prior-updated $\hat{N}_{-1}$. This avoids double-using the initial observation or dropping the first AR(2) likelihood.

3.4 Conditional posteriors of the coefficients and variances

Take the main model as an example, with $\beta = (\alpha, \kappa_0, \delta)'$, $X_t = (a_t, x_t, x_t \bar{N}_t)$, and $y_t^* = y_t - \lambda_{e\zeta}\zeta_t$. Under the independent normal prior $\beta \sim N(b_0, V_0)$, we update jointly from

$$V_1 = \left(V_0^{-1} + \frac{X'X}{\sigma_\eta^2} \right)^{-1}, \tag{18}$$

$$b_1 = V_1 \left(V_0^{-1} b_0 + \frac{X'y^*}{\sigma_\eta^2} \right), \tag{19}$$

$$\beta \mid \mathrm{rest} \sim N(b_1, V_1) \tag{20}$$

The implementation uses the 100-rescaled X and coefficients noted above, but the formula is identical. The baseline estimation imposes no sign restriction, so both δ and the end-of-sample κ_t can be negative.

We update $\lambda_{e\zeta}$ from the normal conditional posterior of the regression of e_t on ζ_t. We update ϕ_1 using not only (7) but also the information in $e_t = \lambda_{e\zeta}(x_t - \phi_1 x_{t-1}) + \eta_t$. We draw ρ_1, ρ_2 from a normal conditional posterior truncated to the AR(2) stationary region,

$$|\rho_2| < 1, \qquad \rho_1 + \rho_2 < 1, \qquad \rho_2 - \rho_1 < 1 \tag{21}$$

The fallback count and rejection rate when no proposal is accepted within 2,000 draws are saved to each run's metadata.

The variance priors are $\sigma^2 \sim \mathrm{IG}(a, b)$ (density kernel $(\sigma^2)^{-(a+1)} \exp(-b/\sigma^2)$), updated from the inverse-gamma conditional posterior that adds the residual sum of squares. The concrete baseline priors are in Table 2.

Table 2: Baseline priors for the main HSA steady model. Normal distributions show mean and standard deviation; inverse-gamma distributions show (a, b). The coefficient priors are common to the other models.

Block	Parameter	Prior	Interpretation
Inflation eq.	α	$N(0.5, 0.2^2)$	Backward inertia
	κ, κ_0	$N(0.1, 0.2^2)$	Slope level
	δ	$N(0, 0.02^2)$	Slope dependence on firm count
	θ, θ_0	$N(0, 0.2^2)$	Cyclical entry effect
	γ	$N(0, 0.02^2)$	Time-varying coefficient of entry
Activity eq.	ϕ_1	$N(0.7, 0.2^2)$	AR(1) of the activity variable
	$\lambda_{e\zeta}$	$N(0, 0.5^2)$	Simultaneous-shock correction
Firm-count state	ρ_1	$N(0.5, 0.2^2)$	Cycle AR(2), truncated to stationary region
	ρ_2	$N(-0.5, 0.2^2)$	Cycle AR(2), truncated to stationary region
	n	$N(0, 0.1^2)$	Trend drift
Variances	σ_η^2	$\mathrm{IG}(2, 2)$	Inflation shock
	σ_ζ^2	$\mathrm{IG}(0.001, 0.001)$	Activity shock
	σ_u^2	$\mathrm{IG}(2, 0.02)$	Cycle shock
	σ_ε^2	$\mathrm{IG}(2, 0.01)$	Trend shock
	σ_N^2	$\mathrm{IG}(2, 0.01)$	Firm-count measurement error

The $\lambda_{e\zeta} \sim N(0, 0.5^2)$ prior and the initial-state prior are code-level defaults shared by the baseline, weak, and tight sweeps; they are not varied by the three YAML prior sets. Table 2 lists the former for completeness, not because it is a baseline-only choice.
For HSA dynamic the shock covariance takes $\Sigma \sim \mathrm{IW}(8, S_0)$ with $S_0 = \mathrm{diag}(3, 3, 0.06, 0.03)$. Under the default (e, ζ)-only, the corresponding marginals of this prior apply to the (e, ζ) block and the two state variances.

3.5 What the Kalman filter and FFBS do

Write the predicted mean and variance at time t as $a_t^s, P_{t|t-1}$ and the filtered mean and variance as m_t, P_t. The forward update for (14)–(16) is

$$a_t^s = c + F m_{t-1}, \qquad P_{t|t-1} = F P_{t-1} F' + Q, \tag{22}$$

$$q_t = z_t - H_t a_t^s, \qquad S_t = H_t P_{t|t-1} H_t' + R, \tag{23}$$

$$K_t = P_{t|t-1} H_t' S_t^{-1}, \qquad m_t = a_t^s + K_t q_t \tag{24}$$

The covariance update uses the numerically stable Joseph form. After drawing the endpoint $s_T \sim N(m_T, P_T)$, we draw the smoothed state path jointly with

$$J_t = P_t F' P_{t+1|t}^{-1}, \tag{25}$$

$$s_t \mid s_{t+1}, z_{1:T} \sim N\big(m_t + J_t(s_{t+1} - c - Fm_t), P_t - J_t P_{t+1|t} J_t'\big) \tag{26}$$

for $t = T-1, \ldots, 0$. Thus the plotted $\bar{N}_t, \hat{N}_t, \kappa_t$ carry the posterior uncertainty not only of the coefficients but also of the state decomposition.

3.6 One Gibbs iteration and the differences across models

One iteration of HSA steady is, in order:

1. update $(\alpha, \kappa_0, \delta)$ from the current $\bar{N}_t$ via (20);
2. update $\lambda_{e\zeta}$ and ϕ_1, and recompute ζ_t, η_t;
3. update $\sigma_\zeta^2, \sigma_\eta^2$;
4. update (ρ_1, ρ_2) and σ_u^2;
5. update n and σ_ε^2, then σ_N^2;
6. update the entire $s_{0:T}$ by FFBS using the updated coefficients and variances.

CES is a conjugate regression benchmark with no latent firm-count state.

HSA dynamic handles the covariance of $r_t = (e_t, \zeta_t, u_t, \varepsilon_t)'$; under the default (e, ζ)-only it updates the 2×2 block of (e_t, ζ_t) and $\sigma_u^2, \sigma_\varepsilon^2$ directly on the constrained parameter space, rather than zeroing the off-diagonal of an unconstrained inverse-Wishart draw after the fact.

HSA full contains the state-by-state product $-\gamma \bar{N}_t \hat{N}_t$, so the joint firm-count state $(\bar{N}_t, \hat{N}_t)$ is not linear-Gaussian at once. HSA full is therefore updated by the joint *Particle Gibbs* step detailed in Section 3.7—a conditional sequential-Monte-Carlo update of the whole state path (512 particles)—which replaces the two alternating exact-FFBS blocks used previously. HSA const-theta ($\gamma = 0$) is a different case: without the bilinear term the inflation observation loads linearly on the joint state, $\big[-\theta_0,\ 0,\ \delta x_t\big] s_t$, so the model is linear-Gaussian and the whole path is drawn in a single exact FFBS sweep—the same routine HSA steady uses, of which HSA steady is the $\theta_0 = 0$ case. In both models the weak signal of the cyclical state $\hat{N}_t$, which must be identified as a separate coefficient, keeps identification weaker than for the other models; neither the joint Particle Gibbs update nor the exact joint FFBS removes that weak identification, and neither is claimed to.

3.7 The Particle Gibbs state update (HSA full)

For HSA full the bilinear term $-\gamma \bar{N}_t \hat{N}_t$ in (6) makes the joint state $s_t = (\hat{N}_t, \hat{N}_{t-1}, \bar{N}_t)'$ non-linear-Gaussian, so the exact FFBS blocks used above do not apply jointly. We instead draw the whole path $s_{0:T}$ with one *Particle Gibbs* sweep—a conditional sequential-Monte-Carlo step that leaves the exact conditional $p(s_{0:T} \mid \theta, y, N^{\text{obs}})$ invariant for any number of particles P (Andrieu, Doucet and Holenstein, 2010).

Proposal and weights. Particles are propagated from the state transition (10)–(11) (a bootstrap proposal): given the ancestor $s_{t-1}^{(i)}$,

$$\hat{N}_t^{(i)} = \rho_1 \hat{N}_{t-1}^{(i)} + \rho_2 \hat{N}_{t-2}^{(i)} + u_t^{(i)}, \qquad \bar{N}_t^{(i)} = n + \bar{N}_{t-1}^{(i)} + \varepsilon_t^{(i)}, \qquad u_t^{(i)} \sim N(0, \sigma_u^2),\ \varepsilon_t^{(i)} \sim N(0, \sigma_\varepsilon^2). \tag{27}$$

Each particle is weighted by *both* observation equations—the inflation equation (6) and the firm-count measurement (9):

$$\log w_t^{(i)} = -\frac{\left(y_t - \mu_t^{(i)}\right)^2}{2\sigma_\eta^2} - \frac{\left(N_t^{\text{obs}} - \bar{N}_t^{(i)} - \hat{N}_t^{(i)}\right)^2}{2\sigma_N^2}, \tag{28}$$

$$\mu_t^{(i)} = \alpha a_t + \left(\kappa_0 + \delta \bar{N}_t^{(i)}\right) x_t - \left(\theta_0 + \gamma \bar{N}_t^{(i)}\right) \hat{N}_t^{(i)} + \lambda_{e\zeta} \zeta_t, \tag{29}$$

with $y_t = \pi_t - \mathbb{E}_t \pi_{t+1}$, $a_t = \pi_{t-1} - \mathbb{E}_t \pi_{t+1}$ and $\zeta_t = x_t - \phi_1 x_{t-1}$ (constants common to all particles at t cancel under normalisation). Normalised weights $W_t^{(i)} = w_t^{(i)} / \sum_j w_t^{(j)}$ are formed with a log-sum-exp for stability, and $\text{ESS}_t = 1/\sum_i (W_t^{(i)})^2$.

Conditional SMC. Let $s_{0:T}^{\text{ref}}$ be the state path from the previous MCMC iteration, pinned to particle slot 1.

1. $t = 0$: set $s_0^{(1)} = s_0^{\text{ref}}$ and draw $s_0^{(i)} \sim N(m_0, P_0)$ for $i \geq 2$ from the initial prior (17); weight by (28).

2. $t = 1, \ldots, T$: resample ancestors $a_t^{(i)} \sim \text{Categorical}(W_{t-1})$ for $i \geq 2$ and set $a_t^{(1)} = 1$, so the reference keeps its lineage; propagate $s_t^{(i)}$ from the transition above given $s_{t-1}^{(a_t^{(i)})}$, with $s_t^{(1)} = s_t^{\text{ref}}$; weight by (28). The pair $(\hat{N}_t, \hat{N}_{t-1})$ in s_t preserves the AR(2) genealogy.

3. Draw a terminal index $J \sim \text{Categorical}(W_T)$ and trace its ancestors backward to return one joint path $s_{0:T}$, which is the new state draw and the reference for the next iteration.

3.8 Iterations, saved draws, and convergence criteria

Each cell uses two independent chains, with a *total* of 12,000 iterations per chain. We discard the first 4,000 as burn-in and save every 5th of the remaining 8,000, giving 1,600 draws per chain and 3,200 draws per cell for diagnostics and summaries. The main analysis is 75 cells: the baseline 45 (3 price indices × 3 activity measures × 5 models) plus the 30 weak/tight-prior cells. Separately, we estimate 2 inverse-markup CES/HSA-dynamic cells with the same settings.

What "converged" means here. The threshold is maximum $\hat{R} \leq 1.01$ and minimum bulk ESS ≥ 400, but a single run has several distinct groups of quantities and they do not pass or fail together. We therefore compute and report the rule on three groups separately:

1. **scalar parameters** — every stored scalar: α, the slope terms, the entry terms, ρ_1, ρ_2, the trend drift n, ϕ_1, $\lambda_{e\zeta}$ and *every* estimated variance;

2. **latent-state paths** — $\bar{N}_t$ and $\hat{N}_t$, maximum $\hat{R}$ and minimum ESS along the path;

3. **derived paths** — κ_t and, where estimated, θ_t.

The † mark in the coefficient tables is the *scalar* rule. A cell marked "OK (coef)" passes on the coefficients but not on the latent-state paths; "OK" passes on all three. We state this explicitly because the two are routinely conflated: marginal coefficient chains can mix well while the joint posterior including the states has not, and reporting only the former would overstate what has been established. Table 23 gives the group-by-group numbers for every model.

Under the main mixed-frequency design, 61 of the 77 runs fail the scalar rule; under the interpolated design the figures are 33 of 77. The two designs fail for different reasons, and the difference is informative rather than incidental. Under interpolation the worst-mixing scalar is the trend drift n in 31 of the 33 failing HSA cells—the level ridge of Section 8 surfacing in the parameter block. Under mixed frequency it is the AR(2) block (ρ_1, ρ_2) in 54 of 61 cells, which for HSA steady is a nuisance object because $\theta = 0$ keeps $\hat{N}_t$ out of the inflation equation. The parameters the conclusions rest on fail far less often than the run as a whole, under either design: δ in 10 of 61 cells and κ_0 in 9 under mixed frequency, against 5 and 3 under interpolation. The 95% credible intervals are the 2.5% and 97.5% quantiles of the saved draws; the plotted points are posterior means.

4 Main results

All results in this section use the **mixed-frequency** firm-count design: the 31 annual observations enter in their own Q4 and the 93 intervening quarters are inferred inside the state-space model (Section 3.3). Table 3 gives the main-model (HSA steady) estimates for every price index and activity measure, so each result can be read against the specification it comes from. The main specification—**core CPI, negative unemployment gap**—is in bold; a positive δ means the slope flattens as the firm count falls.

Table 3: Headline results (main model: HSA steady, baseline priors, mixed-frequency annual-Q4, $T = 124$), by price index and activity measure. δ is the competition dependence of the slope (a positive δ means the slope flattens as the firm count falls); BF_{10} is the Savage–Dickey Bayes factor against $\delta = 0$; κ_0 is the slope at average competition; the last column is the implied κ_t from the start to the end of the sample. The **main specification (core CPI, negative unemployment gap)** is in bold. †: fails the *coefficient* convergence rule ($\hat{R} \leq 1.01$ and bulk ESS ≥ 400 over every scalar parameter, including the trend drift n and all variances). "OK (coef)" marks a cell that passes on the coefficients but not on the latent-state paths; the group-by-group diagnostics are in Table 23. The δ $\hat{R}$/ESS column is that coefficient's own mixing, which the blanket flag does not show.

Activity measure	Price	δ [95% CI]	BF_{10}	κ_0 [95% CI]	κ_t start→end	δ $\hat{R}$/ESS	Conv
Negative unemployment gap	**Core CPI**	**+0.026 [+0.012, +0.041]**†	**116.7**	**+0.058 [+0.023, +0.092]**	**+0.126 → -0.007**	**1.000 / 2986**	**watc**
	Headline CPI	+0.031 [+0.007, +0.056]†	13.5	+0.078 [+0.012, +0.145]	+0.141 → -0.004	1.000 / 2641	watc
	PPI	+0.010 [-0.027, +0.046]†	1.1	+0.087 [-0.119, +0.291]	+0.104 → +0.063	1.001 / 3134	watc
HP output gap	Core CPI	+0.026 [+0.000, +0.048]†	11.1	+0.077 [+0.024, +0.128]	+0.149 → +0.008	1.002 / 663	watc
	Headline CPI	-0.005 [-0.037, +0.027]†	0.9	+0.139 [+0.047, +0.233]	+0.134 → +0.153	1.000 / 2899	watc
	PPI	-0.003 [-0.042, +0.033]†	1.0	+0.209 [-0.051, +0.473]	+0.205 → +0.217	1.000 / 2882	watc
BN output gap	Core CPI	-0.001 [-0.033, +0.025]†	0.7	+0.029 [-0.022, +0.080]	+0.038 → +0.033	1.016 / 143	watc
	Headline CPI	-0.018 [-0.050, +0.014]†	1.6	+0.055 [-0.032, +0.138]	+0.033 → +0.102	1.001 / 2713	watc
	PPI	-0.006 [-0.045, +0.033]†	1.1	+0.188 [-0.063, +0.436]	+0.177 → +0.205	1.001 / 3270	watc

Reading the convergence column. Every cell in Table 3 carries a †, and it would be a mistake to read that as "δ is not estimated reliably here." The blanket flag is set by whichever scalar mixes worst, and under mixed frequency that is the AR(2) block (ρ_1, ρ_2) in 54 of the 61 HSA cells. For **HSA steady** the AR(2) block governs $\hat{N}_t$, which—because $\theta = 0$ in this model—does not enter the inflation equation at all. The coefficient the conclusions rest on is δ, and its own diagnostics are in the penultimate column: δ alone clears $\hat{R} \leq 1.01$ and bulk ESS ≥ 400 in 8 of the 9 HSA-steady baseline cells, at a median bulk ESS of 2,882. That is a statement about one coefficient, not about the run: no HSA cell in this design passes the whole-run rule, which is applied to all 18 scalars at once. Section 8 states which failures do and do not bear on the conclusions.

Table 4 then places the main model against the other model variants (CES, HSA dynamic, HSA const-theta, HSA full) for the negative unemployment gap. CES and HSA dynamic hold the slope constant, so they have no δ; the HSA models with a firm-count-dependent slope estimate δ, and HSA steady is the main model. Full coefficients (θ, γ, and the κ_t path) for every activity measure are in Appendix C.

Table 4: Model comparison for the negative unemployment gap (baseline priors, mixed-frequency annual-Q4), by price index. Slope is κ for CES / HSA dynamic and κ_0 otherwise; the competition dependence δ (and its BF_{10}) is defined only for the HSA models with a firm-count-dependent slope. The main model, **HSA steady**, is in bold; full coefficients (θ, γ, κ_t path) for every activity measure are in Appendix C. State blocks: CES conjugate; HSA steady / HSA dynamic exact joint FFBS; HSA const-theta joint FFBS; HSA full Particle Gibbs. †: fails the coefficient convergence rule; "OK (coef)" passes on coefficients but not on the latent-state paths.

Model	Price	slope [95% CI]	δ [95% CI]	$BF_{10}(\delta)$	Conv.
HSA steady	**Core CPI**	**+0.058 [+0.023, +0.092]**	**+0.026 [+0.012, +0.041]†**	**116.7**	**watch**
	Headline CPI	**+0.078 [+0.012, +0.145]**	**+0.031 [+0.007, +0.056]†**	**13.5**	**watch**
	PPI	**+0.087 [-0.119, +0.291]**	**+0.010 [-0.027, +0.046]†**	**1.1**	**watch**
HSA dynamic	Core CPI	+0.038 [+0.007, +0.069]	–	–	watch
	Headline CPI	+0.054 [-0.010, +0.116]	–	–	watch
	PPI	+0.057 [-0.164, +0.274]	–	–	watch
HSA const-theta	Core CPI	+0.062 [+0.025, +0.105]	+0.027 [+0.010, +0.045]†	40.2	watch
	Headline CPI	+0.080 [+0.014, +0.145]	+0.031 [+0.006, +0.056]†	11.5	watch
	PPI	+0.087 [-0.126, +0.300]	+0.010 [-0.028, +0.048]†	1.1	watch
HSA full	Core CPI	+0.068 [+0.023, +0.113]	+0.030 [+0.013, +0.048]†	119.8	watch
	Headline CPI	+0.079 [+0.006, +0.151]	+0.030 [+0.002, +0.055]†	7.6	watch
	PPI	+0.087 [-0.123, +0.290]	+0.009 [-0.028, +0.047]†	1.1	watch

4.1 The negative-unemployment-gap slope flattens as the firm count falls

For the baseline HSA steady on the main sample, the core-CPI negative-unemployment-gap specification gives

$$\delta = +0.026[+0.012, +0.041],$$

and the Savage–Dickey Bayes factor against $\delta = 0$ is $BF_{10} = 116.7$. Because $\bar{N}_t$ falls over the sample, the implied slope κ_t declines monotonically from $+0.126$ to -0.007: as the firm count falls (concentration rises), the Phillips curve flattens, the sign the HSA mechanism predicts. Headline CPI also gives $\delta = +0.031[+0.007, +0.056]$ ($BF_{10} = 13.5$), so the CPI result for the negative unemployment gap is robust to the inflation index. Figure 2 plots the implied κ_t path; its end-of-sample band includes zero, so the accurate statement is a flattening to near zero rather than a strictly positive residual slope.

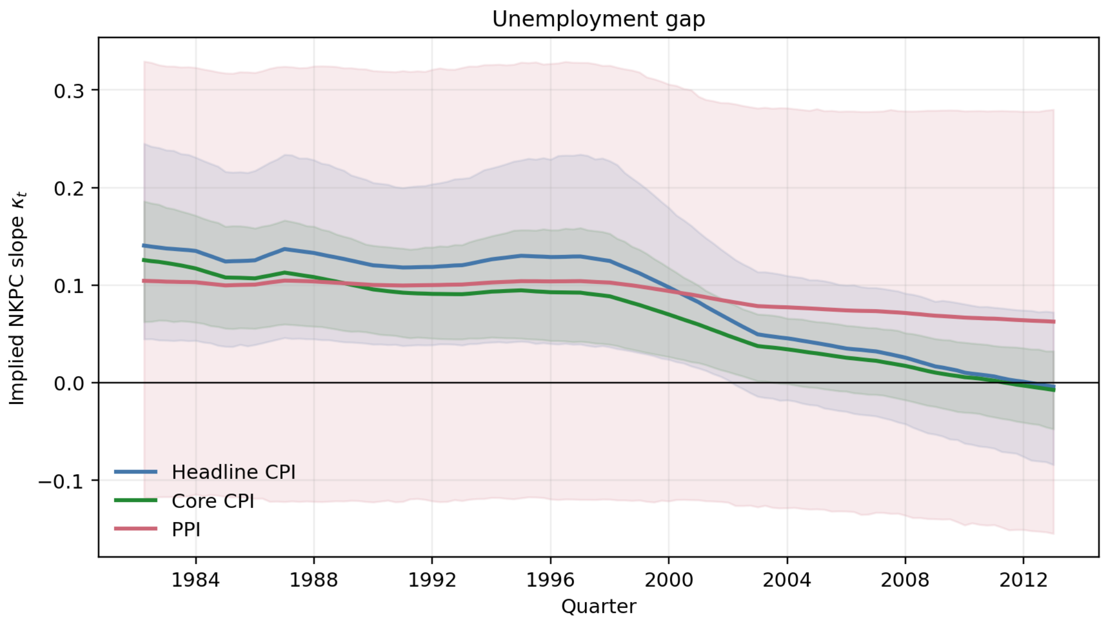

Figure 2: Time-varying slope κ_t for the negative unemployment gap under the mixed-frequency design (posterior mean and 95% band). The band includes the uncertainty from inferring the 93 unobserved quarterly firm counts. Under CPI it declines to near zero by the end of the sample, whereas the PPI band is wide.

The estimated Phillips-curve slope in the core-CPI specification declines from +0.126 at the beginning of the sample to approximately zero by the end. This change is economically meaningful. For example, a four-point deterioration in the negative unemployment gap ($x_t = -4$) implies an inflation response of about −0.50 percentage point at the 1982 slope, but almost no response at the 2012 slope.

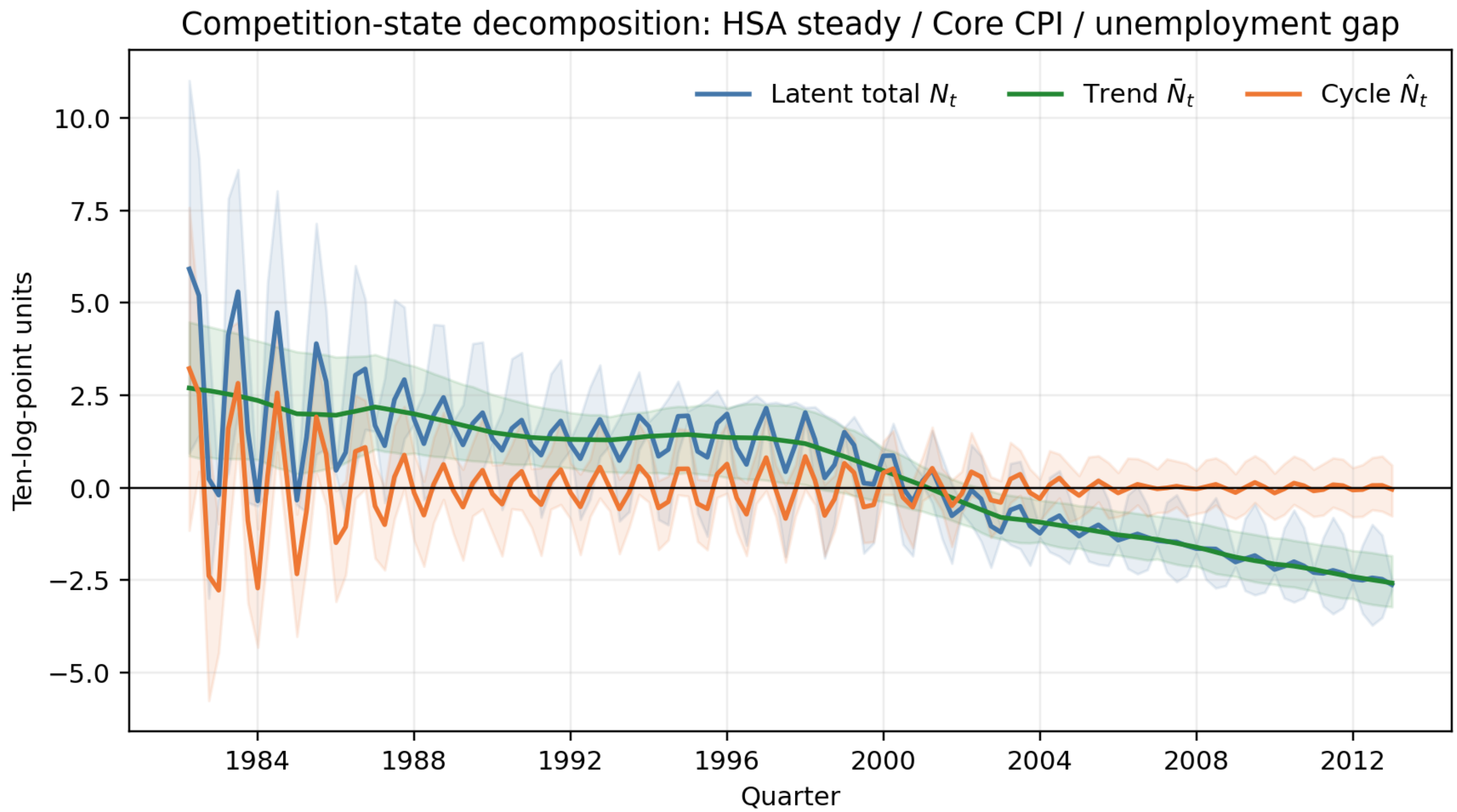

Figure 3: Posterior decomposition of the firm count $N_t = \bar{N}_t + \hat{N}_t$ under the main HSA steady (negative unemployment gap, core CPI), mixed-frequency design. Lines are posterior means, bands are 95% credible intervals. The quarterly states between the Q4 annual observations are inferred, not interpolated, so the bands widen between anchors.

5 The CES slope bias and the constant-slope benchmark

5.1 Theoretical prediction and its empirical counterpart

Fujiwara–Matsuyama (2026) show that because endogenous entry generates a cost-push term through strategic complementarity, a CES-type equation that regresses inflation on real marginal cost alone under-estimates the NKPC slope. In our HSA-dynamic representation, moving expected inflation to the left-hand side gives

$$y_t = \alpha a_t + \kappa_{\mathrm{HSA}} x_t + q_t + e_t, \qquad q_t = -\theta \hat{N}_t, \tag{30}$$

where $y_t = \pi_t - \mathbb{E}_t \pi_{t+1}$ and $a_t = \pi_{t-1} - \mathbb{E}_t \pi_{t+1}$. When the CES equation omits q_t,

$$\mathrm{Bias}(\hat{\kappa}_{\mathrm{CES}}) = \mathrm{plim}(\hat{\kappa}_{\mathrm{CES}} - \hat{\kappa}_{\mathrm{HSA}}) = \frac{\tilde{x}'\tilde{q}}{\tilde{x}'\tilde{x}}, \tag{31}$$

where tildes denote residuals after removing a_t and the activity shock $\zeta_t = x_t - \phi_1 x_{t-1}$. The theory predicts (31) is negative, i.e. $\kappa_{\mathrm{CES}} - \kappa_{\mathrm{HSA}} < 0$.

To keep this correspondence sharp, separately from the slack measures of the main results we use the BN-decomposed inverse markup as a proxy for real marginal cost and re-estimate CES and HSA dynamic with the same sample, priors, and MCMC settings. For each HSA posterior draw we form $q_t^{(s)} = -\theta^{(s)} \hat{N}_t^{(s)}$ and directly compute the bias in (31). Because the CES–HSA difference is a comparison across separately estimated posteriors, we build a Monte Carlo distribution of the difference by randomly permuting the HSA draws with a fixed seed.

5.2 Estimation results

For the inverse-markup specification that corresponds directly to the theory,

$$\begin{aligned} \kappa_{\mathrm{CES}} &= +0.007[-0.217, +0.228], & \kappa_{\mathrm{HSA}} &= +0.009[-0.208, +0.231], \\ \kappa_{\mathrm{CES}} - \kappa_{\mathrm{HSA}} &= -0.003[-0.308, +0.306], & \Pr(\kappa_{\mathrm{CES}} - \kappa_{\mathrm{HSA}} < 0) &= 0.510, \\ B_{\mathrm{HSA}} &= +0.000[-0.013, +0.012], & \Pr(B_{\mathrm{HSA}} < 0) &= 0.475 \end{aligned}$$

The CES–HSA difference does not show a negative bias. The omitted-variable bias implied by HSA itself also has a point estimate of the sign opposite to the theory (positive), with a credible interval that includes zero.

Table 5: CES–HSA-dynamic slope comparison and the omitted-variable bias implied by HSA. Each interval is a 95% credible interval. † marks that at least one of CES or HSA dynamic is outside the convergence criteria.

specification	CES κ	HSA κ	CES-HSA	Pr(CES-HSA<0)	HSA-implied OVB	Pr(OVB<0)
Inverse markup (theory)	+0.007 [-0.217, +0.228]†	+0.009 [-0.208, +0.231]†	-0.003 [-0.308, +0.306]†	0.510	+0.000 [-0.013, +0.012]†	0.475
Headline CPI / negative unemployment gap	+0.054 [-0.009, +0.115]†	+0.054 [-0.010, +0.116]†	+0.000 [-0.089, +0.091]†	0.499	+0.001 [-0.002, +0.005]†	0.429
Core CPI / negative unemployment gap	+0.039 [+0.009, +0.070]†	+0.038 [+0.007, +0.069]†	+0.001 [-0.042, +0.043]†	0.484	+0.001 [-0.001, +0.016]†	0.349
PPI / negative unemployment gap	+0.082 [-0.118, +0.288]†	+0.057 [-0.164, +0.274]†	+0.025 [-0.267, +0.325]†	0.436	+0.000 [-0.004, +0.005]†	0.415

Verdict. The inverse-markup row passes the convergence criteria for both models, but the slope intervals are wide and neither the CES–HSA difference nor the within-model OVB supports the negative sign the theory predicts. The three auxiliary rows using the negative unemployment gap give the same conclusion. This does not amount to a refutation of the theory: the inverse markup is an aggregate series rather than firm-level real marginal cost, and HSA dynamic is not a joint estimate of the full theoretical model. What we can say from this paper is only that *in the current data and semi-structural model, we do not obtain an estimate corresponding to the theory paper's prediction of a negative bias in the CES slope.*

As a benchmark for the time-varying result of Section 4, we also estimate the constant-slope models. The full model-by-model coefficient tables for the negative unemployment gap, and the corresponding prior-sensitivity table, are collected in Appendix C.

CES vs HSA (constant κ) CES is the constant-slope benchmark. Among the converged cells, the core-CPI κ interval sits on the positive side, while headline CPI and PPI include zero. CES measures the existence of the average curve but does not explain the flattening. The prior-vs-posterior of the slope (Figure 4) shows CES's constant κ and HSA steady's slope at average competition κ_0, both sharply updated by the data and sitting just above zero.

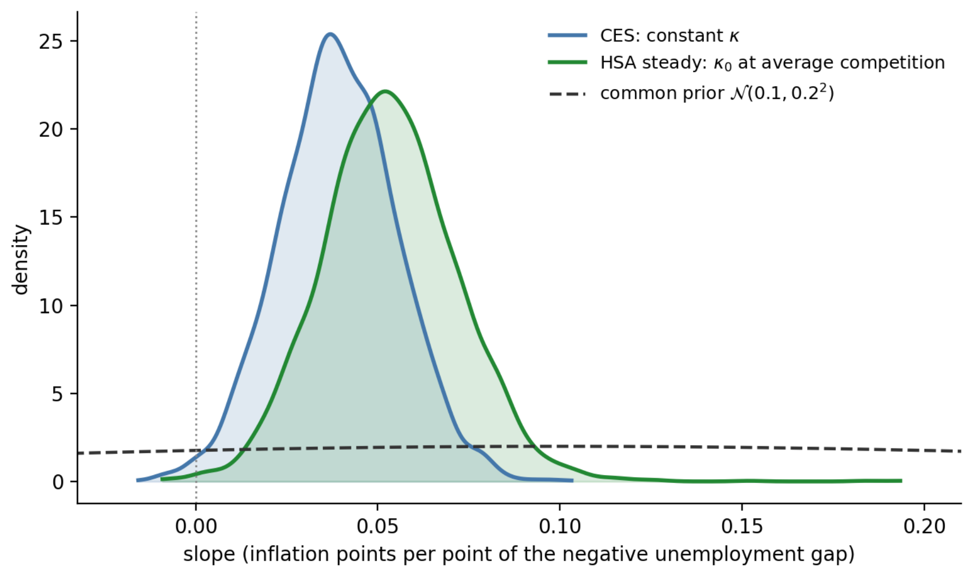

Figure 4: Prior vs. posterior of the Phillips-curve slope for the negative unemployment gap: CES's constant slope κ and HSA steady's slope at average competition κ_0, against their common prior $\mathcal{N}(0.1, 0.2)$. Both are sharply updated by the data and sit just above zero; the two models agree on the slope level, so what distinguishes them is whether that slope varies with the firm count (Section 6).

6 Does the time-varying slope improve fit over the constant slope?

We compare CES (constant κ) against HSA steady (slope varying with the firm count) by two criteria: predictive fit (Table 6) and the Savage–Dickey Bayes factor for $\delta \neq 0$ (Table 7).

Fit, by four scores at once. Earlier revisions reported only an *in-sample plug-in* score. It evaluates the inflation equation at posterior-mean parameters and states (bars denote posterior means),

$$\hat{\pi}_t = \bar{\alpha}\,\pi_{t-1} + (1-\bar{\alpha})\,\mathbb{E}_t\pi_{t+1} + \bar{\kappa}_t\,x_t, \qquad \bar{\kappa}_t = \bar{\kappa}_0 + \bar{\delta}\,\bar{N}_t \;\; (\text{CES: } \bar{\kappa}_t = \bar{\kappa}), \tag{32}$$

and scores the residuals $r_t = \pi_t - \hat{\pi}_t$ by a Gaussian log-likelihood at the posterior-mean shock variance,

$$S = -\frac{T}{2}\log\bigl(2\pi\bar{\sigma}_e^2\bigr) - \frac{1}{2\bar{\sigma}_e^2}\sum_{t=1}^{T} r_t^2. \tag{33}$$

This reuses the estimation sample and collapses the posterior to a point, so it is neither an out-of-sample forecast nor a posterior predictive density.

We now report it *alongside* three proper scores rather than in place of them: the prequential one-step-ahead log predictive density $\mathrm{LPD}_1 = \sum_t \log p(\pi_t \mid \pi_{1:t-1}, x, N)$, integrated over the posterior; WAIC; and PSIS-LOO. Keeping the plug-in score visible is the point of the table—it lets the reader see how much a crude criterion overstates the case:

Table 6: Fit comparison for the negative unemployment gap, mixed-frequency design. All four columns are log scores, so higher is better, and all are differences against nothing—only comparisons within a price index are meaningful. *Plug-in* is the in-sample score of the text: the inflation equation evaluated at posterior-mean parameters and states, scored by a Gaussian log-likelihood at the posterior-mean shock variance. LPD_1 is the prequential one-step-ahead log predictive density, $\sum_t \log p(\pi_t \mid \pi_{1:t-1}, x, N)$, integrated over the posterior. WAIC and PSIS-LOO are computed on the pointwise inflation log-likelihood. $\hat{k}$ is the maximum Pareto shape for the LOO importance weights; $\hat{k} > 0.7$ marks a cell where LOO is unreliable.

Price	Model	Plug-in	LPD_1	WAIC	PSIS-LOO	$\hat{k}$
Core CPI	CES	-17.98	-17.16	-22.43	-22.53*	0.75
	HSA steady	**-8.53**	**-12.34**	**-18.60**	**-18.61**	**0.60**
	HSA dynamic	-16.14	-14.42	-21.11	-21.18*	0.97
	HSA full	-8.33	-8.81	-17.76	-17.69	0.52
Headline CPI	CES	-119.90	-116.43	-124.39	-124.29	0.60
	HSA steady	**-115.08**	**-113.45**	**-122.09**	**-122.06**	**0.66**
	HSA dynamic	-119.74	-118.14	-125.33	-125.22	0.64
	HSA full	-115.53	-113.55	-122.34	-122.39*	0.71
PPI	CES	-294.19	-288.76	-298.18	-298.38*	1.19
	HSA steady	**-293.97**	**-288.33**	**-297.95**	**-297.56**	**0.69**
	HSA dynamic	-294.18	-291.68	-301.18	-301.03*	0.91
	HSA full	-293.99	-288.40	-298.71	-298.45*	0.87

For HSA steady against CES the plug-in score gives +9.45 on core CPI where LPD_1 gives +4.82 and WAIC and LOO give +3.83 and +3.92; on headline CPI it gives +4.82 against +2.98, +2.30 and +2.23. The plug-in score overstates the improvement by a factor of roughly 1.6 to 2.0 against the prequential score, which is what one expects from a criterion that rewards a model for the extra latent states it fitted to this same sample. The *ordering* nevertheless survives: on both CPI indices the time-varying slope improves every score, on PPI no score meaningfully separates the models (LPD_1 +0.43, WAIC +0.23, LOO +0.82), and HSA dynamic—the entry channel without a time-varying slope—never beats HSA steady. Two caveats. First, the LOO column is unreliable wherever the maximum Pareto $\hat{k}$ exceeds 0.7, which is 6 of the 12 cells (the $\hat{k}$ column of Table 6 marks which). That includes the CES baseline for the core-CPI comparison, so the +3.92 LOO difference quoted above rests on one number that PSIS itself flags; the WAIC difference of +3.83, which does not depend on importance sampling, is the more trustworthy of the two, and it agrees. The PPI cells should simply not be read off the LOO column. Second, all four scores are computed on the estimation sample—LPD_1 is prequential in the sense that each π_t is scored against a predictive density formed without π_t, but the parameters were estimated on the full sample—so none of this is a genuine out-of-sample exercise.

Bayes factor (Savage–Dickey), stated precisely. We test $\delta = 0$ *within the HSA state model*: M_1 is HSA steady with δ free and M_0 is the same model with $\delta = 0$. The Savage–Dickey identity gives

$$\mathrm{BF}_{10} = \frac{p(\delta = 0)}{\pi(\delta = 0 \mid y)}, \qquad p(\delta = 0) = \frac{1}{\sigma_\delta\sqrt{2\pi}} = \frac{1}{0.02\sqrt{2\pi}} \approx 19.95 \;\; (\delta \sim \mathcal{N}(0, \sigma_\delta^2)), \tag{34}$$

The posterior ordinate is computed *without density estimation.* The Gibbs update for the inflation coefficients is an exact Gaussian conditional, so the marginal ordinate is its Rao–Blackwellised average over the retained draws of everything else, $\psi = (\bar{N}_{0:T}, \hat{N}_{0:T}, \sigma_\eta^2, \lambda_{e\zeta}, \phi_1)$:

$$\hat{\pi}(\delta = 0 \mid y) = \frac{1}{S}\sum_{s=1}^{S} \mathcal{N}\big(0;\; b_1^{(s)}[\delta],\, V_1^{(s)}[\delta,\delta]\big), \qquad \psi^{(s)} \sim p(\psi \mid y), \tag{35}$$

with $b_1^{(s)}, V_1^{(s)}$ the conditional moments of (20) evaluated at $\psi^{(s)}$. This matters because $\delta = 0$ lies about three posterior standard deviations outside the sampled range—no retained draw of δ is negative—so a kernel density evaluated there extrapolates its own tail rather than measuring the posterior: a Gaussian kernel estimate at that point moves by orders of magnitude as its bandwidth is scaled around Scott's rule, whereas (35) draws on 316 effectively contributing terms out of 3,200 (the three largest carry 4.7% of the sum) and has a Monte Carlo standard error of about 5%.

Three cautions remain. (i) This is *not* a CES-vs-HSA marginal-likelihood ratio: CES and HSA also differ in the latent-N block, which the SDDR holds fixed, so we keep the comparison inside the HSA state model. (ii) The theory is one-sided ($\delta > 0$); the two-sided BF_{10} is conservative. (iii) A Bayes factor against a point null is proportional to the prior width by construction, so these numbers are statements about $\delta = 0$ under the stated $\mathcal{N}(0, 0.02^2)$ prior and are not prior-free.

Table 7: Constant vs. time-varying slope: is $\delta \neq 0$? Posterior mean of δ [95% CI] with the Savage–Dickey Bayes factor BF_{10} for $\delta \neq 0$ within the HSA state model in parentheses (HSA steady, baseline priors, **mixed-frequency** design, full sample, unrestricted). $\mathrm{BF}_{10} > 1$ favors a concentration-dependent slope.

activity	Headline CPI	Core CPI	PPI
Negative unemployment gap	+0.031 [+0.007, +0.056] (13.5)	+0.026 [+0.012, +0.041] (116.7)	+0.010 [-0.027, +0.046] (1.1)
HP output gap	-0.005 [-0.037, +0.027] (0.9)	+0.026 [+0.000, +0.048] (11.1)	-0.003 [-0.042, +0.033] (1.0)
BN output gap	-0.018 [-0.050, +0.014] (1.6)	-0.001 [-0.033, +0.025] (0.7)	-0.006 [-0.045, +0.033] (1.1)

Verdict. For the negative unemployment gap, allowing the slope to vary with the firm count improves the in-sample fit score and the Bayes factor favors $\delta \neq 0$ strongly under both CPI indices. For the headline output gaps the gain is marginal, consistent with the appendix tables: a positive competition dependence appears only in the core-CPI HP specification, not in BN. For the labor-share gap and inverse markup there is no inflation signal, so constant and time-varying slopes fit equally well. We stress that this is an in-sample fit comparison, not evidence of out-of-sample predictive gains.

Limits of the Savage–Dickey ratios. We read them only as the point null $\delta = 0$ *within the HSA state model*; they are not a CES-vs-HSA marginal-likelihood ratio. The in-sample LPPD and RMSE conditioned on the latent states are likewise not out-of-sample predictions, so they are not used as grounds to select a more complex model. The model-versus-model question is answered separately, below.

6.1 Conditional marginal likelihood: CES versus HSA steady

The Savage–Dickey ratio asks whether $\delta = 0$ inside one model; the fit scores ask which model tracks this sample better. Neither is a model comparison in the Bayesian sense. That requires the two models to be densities of the same data given the same conditioning set,

$$\log p(\pi \mid x, N^{\text{obs}}, M) = \log m(\pi, N^{\text{obs}}, x \mid M) - \log m(N^{\text{obs}} \mid M) - \log m(x), \tag{36}$$

with the latent states integrated out exactly by the Kalman filter and the static parameters integrated out by Chib's (1995) identity, using the production sampler's own block conditionals and explicit reduced Gibbs runs.

The two subtracted terms matter. The Gibbs posterior conditions on π, N^{obs} *and* x: σ_ζ^2 is drawn from the activity-equation residuals, and both ϕ_1 and σ_ζ^2 are ordinate blocks. So the conditioning set is (x, N^{obs}), and both Occam factors have to be refunded rather than charged. They separate because $p(N \mid \theta)$ touches only the firm-count block and $p(x \mid \phi_1, \sigma_\zeta^2)$ only the activity block. $\log m(x)$ is the same number for both models; we compute it once per cell and debit both with it, rather than cancelling it by assertion, so that Table 8 shows the two models conditioned on the identical quantity. CES has no firm-count block and therefore no $\log m(N)$ term.

Table 8: Conditional marginal likelihood, CES versus HSA steady, mixed-frequency design, baseline priors. Each row is $\log p(\pi \mid x, N^{\text{obs}}, M)$ for the two models and their difference. The two subtracted columns are Occam factors for data the comparison conditions on: $\log m(N^{\text{obs}})$ for the firm-count block, which CES does not have, and $\log m(x)$ for the activity equation, which is identical for the two models and is computed once per cell rather than assumed to cancel. BF > 1 favours HSA steady. The main cell is in bold and is the only one run at three seeds; its Monte Carlo standard error is in the text.

Price	Activity	$\log m(\pi, N, x)$ HSA	$\log m(N)$ HSA	$\log m(\pi, x)$ CES	$\log m(x)$	HSA cond.	CES cond.	BF
Headline CPI	Unemployment gap	-192.8	-24.7	-171.0	-45.3	-122.8	-125.7	17.1
	HP output gap	-264.0	-24.7	-239.3	-118.5	-120.8	-120.8	1.0
	BN output gap	-269.4	-24.7	-245.4	-117.2	-127.4	-128.1	2.0
Core CPI	**Unemployment gap**	**-103.9**	**-24.5**	**-83.9**	**-45.3**	**-34.0**	**-38.6**	**93.9**
	HP output gap	-175.2	-24.7	-153.4	-118.5	-32.0	-34.8	16.7
	BN output gap	-181.1	-24.7	-156.2	-117.2	-39.2	-39.0	0.8
PPI	Unemployment gap	-369.1	-24.7	-344.7	-45.3	-299.1	-299.4	1.3
	HP output gap	-439.7	-24.7	-415.0	-118.5	-296.5	-296.4	1.0
	BN output gap	-440.4	-24.7	-415.9	-117.2	-298.5	-298.6	1.1

For the main cell the comparison favours HSA steady by $\Delta \log m = +4.52$, a Bayes factor of about 92. Across 3 seeds the Monte Carlo standard error is 0.16, which moves the Bayes factor within 67–115 without touching its sign or its evidence class. That uncertainty is almost entirely the firm-count block: the standard deviation of $\log m(N)$ across seeds is 0.225 against 0.002 for the CES term, the same slow mixing of the AR(2) state that Section 8 describes, now showing up in the reduced runs rather than in the posterior.

The pattern across cells reproduces, by a different route, what δ already said. The core-CPI negative-unemployment-gap cell is the strongest; core CPI with the HP output gap and headline CPI with the unemployment gap are moderate; the BN output gap and all three PPI cells are indistinguishable from CES. This is an independent check rather than a restatement: it compares models rather than testing a coefficient, and it is computed from the marginal likelihood rather than from the posterior of δ.

Two caveats. The estimator is exact only for the linear-Gaussian models: **HSA full** is not linear-Gaussian in the joint state, so it has no exact conditional likelihood and no Gibbs ordinate over its blocks, and the implementation raises rather than returning a posterior-mean plug-in. And Chib's estimator carries no internal warning when one draw dominates an ordinate factor; ours refuses to report a factor supported by too few effective draws, which is how a 660-log-point error on one seed was caught rather than published. 0 of the reported cells hit that guard.

7 What PCHIP interpolation does to the estimates

The conventional alternative to the mixed-frequency design of Section 3.3 is to interpolate the annual firm count to a quarterly series and then use that series *as data*. This section reports that version and documents what the interpolation costs. Everything else—3 price indices, 3 activity measures, 5 models, 3 priors, the inverse-markup slope-bias diagnostic, and all MCMC settings—is identical. CES contains neither N_t nor a latent firm-count state, so its likelihood and posterior are invariant to the observation scheme and its 16 cells are shared across both designs.

7.1 The two designs, side by side

Setting $\mathcal{T}_N = \{1, \ldots, T\}$ in (12)—every quarter observed—recovers the interpolated filter exactly; the mixed-frequency design is that same filter with 93 of the 124 firm-count rows removed. The estimator is not the issue. What differs is what is fed to it.

Table 9: What the interpolation changes. Posterior quantities for the main cell (HSA steady, core CPI, negative unemployment gap, baseline priors).

	Mixed-frequency (main)	PCHIP-interpolated
Firm-count observations used	31 of 124	124 of 124
Estimated measurement error σ_N	0.069	0.023
ρ_1, ρ_2	+0.19, −0.89	+1.80, −0.81
Implied AR(2) period	4.3 quarters	94.0 quarters
Mean reversion $1 - \rho_1 - \rho_2$	1.698	0.014
Posterior corr$(\bar{N}_0, \hat{N}_0)$	+0.1957	−0.9996
Posterior corr$(\bar{N}\text{ level}, \kappa_0)$	−0.35	−0.52
δ	+0.026 [+0.012, +0.041]	+0.025 [+0.012, +0.042]
κ_t start $\to$ end	+0.126 $\to$ −0.007	+0.132 $\to$ −0.010

Three things follow. First, the interpolated series is so smooth that the model drives σ_N down to 0.023 to fit it, which pins $\bar{N}_t + \hat{N}_t$ at all 124 quarters and leaves only the split free. Second, the AR(2) cycle is pushed to a near-unit root, $\rho_1 + \rho_2 = 0.986$, so its mean-reversion term $1 - \rho_1 - \rho_2$ collapses to 0.014. That term is the *only* thing anchoring the level of $\hat{N}_t$: adding a constant c to $\bar{N}$ and subtracting it from $\hat{N}$ leaves the firm-count equation, the trend equation and—through $\kappa_0 \to \kappa_0 - \delta c$—the inflation equation all exactly unchanged. When the anchor vanishes the two states become almost perfectly counter-identified, which is the −0.9996 in the table. Across all 122 estimated HSA cells the correlation between $1 - \rho_1 - \rho_2$ and corr$(\bar{N}_0, \hat{N}_0)$ is +0.93, so this is the mechanism and not a coincidence. Third, the "cycle" the interpolated design recovers has a 94.0-quarter period: it is a second trend competing with $\bar{N}_t$ for the same low-frequency variation, not a business-cycle object.

What is *not* affected. δ is essentially unchanged—the interaction $x_t \bar{N}_t$ that identifies it is insensitive to a level shift in $\bar{N}$, which is absorbed by κ_0. Across the nine HSA-steady baseline cells the two designs agree on δ to the third decimal in eight of them. The substantive conclusion of this paper does not depend on the choice; the reported precision and the interpretation of the latent states do.

7.2 Interpolated results, for comparison

Table 10: Headline results (main model: HSA steady, baseline priors, PCHIP-interpolated, $T = 124$), by price index and activity measure. δ is the competition dependence of the slope (a positive δ means the slope flattens as the firm count falls); BF_{10} is the Savage–Dickey Bayes factor against $\delta = 0$; κ_0 is the slope at average competition; the last column is the implied κ_t from the start to the end of the sample. The **main specification (core CPI, negative unemployment gap)** is in bold. †: fails the *coefficient* convergence rule ($\hat{R} \leq 1.01$ and bulk ESS ≥ 400 over every scalar parameter, including the trend drift n and all variances). "OK (coef)" marks a cell that passes on the coefficients but not on the latent-state paths; the group-by-group diagnostics are in Table 14. The δ $\hat{R}$/ESS column is that coefficient's own mixing, which the blanket flag does not show.

Activity measure	Price	δ [95% CI]	BF_{10}	κ_0 [95% CI]	κ_t start→end	δ $\hat{R}$/ESS	Conv.
Negative unemployment gap	**Core CPI**	**+0.025 [+0.012, +0.042]**	**464.7**	**+0.053 [+0.018, +0.090]**	**+0.132 → -0.010**	**1.000 / 1743**	**OK**
	Headline CPI	+0.031 [+0.010, +0.053]	36.2	+0.067 [+0.004, +0.131]	+0.172 → -0.016	1.001 / 2718	OK
	PPI	+0.012 [-0.026, +0.049]†	1.2	+0.087 [-0.117, +0.287]	+0.124 → +0.056	1.000 / 3131	watch
HP output gap	Core CPI	+0.029 [+0.011, +0.048]	66.7	+0.069 [+0.016, +0.120]	+0.162 → -0.006	1.000 / 2213	OK
	Headline CPI	-0.002 [-0.032, +0.028]	0.7	+0.143 [+0.046, +0.237]	+0.136 → +0.148	1.000 / 3159	OK
	PPI	-0.004 [-0.043, +0.033]	1.0	+0.212 [-0.066, +0.484]	+0.198 → +0.224	1.000 / 3039	OK
BN output gap	Core CPI	+0.011 [-0.008, +0.030]	0.9	+0.030 [-0.021, +0.081]	+0.063 → +0.004	1.001 / 2969	OK
	Headline CPI	-0.013 [-0.042, +0.016]	1.1	+0.065 [-0.019, +0.146]	+0.023 → +0.098	1.000 / 3228	OK
	PPI	-0.008 [-0.045, +0.030]	1.1	+0.190 [-0.058, +0.429]	+0.164 → +0.211	1.000 / 3218	OK

Table 11: Interpolated design: model-by-model results for the negative unemployment gap.

model	inflation	slope	δ	$BF_{10}(\delta)$	entry	γ	κ_t path
CES	Headline CPI	+0.054 [-0.009, +0.115]	–	–	–	–	–
CES	Core CPI	+0.039 [+0.009, +0.070]	–	–	–	–	–
CES	PPI	+0.082 [-0.118, +0.288]	–	–	–	–	–
HSA steady	Headline CPI	+0.067 [+0.004, +0.131]	+0.031 [+0.010, +0.053]	36.2	–	–	+0.172 → -0.016
HSA steady	Core CPI	+0.053 [+0.018, +0.090]	+0.025 [+0.012, +0.042]	464.7	–	–	+0.132 → -0.010
HSA steady	PPI	+0.087 [-0.117, +0.287]	+0.012 [-0.026, +0.049]	1.2	–	–	+0.124 → +0.056
HSA dynamic	Headline CPI	+0.052 [-0.025, +0.144]	–	–	-0.014 [-0.150, +0.162]	–	–
HSA dynamic	Core CPI	+0.032 [-0.003, +0.067]	–	–	-0.025 [-0.077, +0.033]	–	–
HSA dynamic	PPI	+0.050 [-0.165, +0.269]	–	–	-0.032 [-0.338, +0.341]	–	–
HSA const-theta	Headline CPI	+0.077 [-0.001, +0.151]	+0.035 [+0.009, +0.061]	20.4	+0.034 [-0.082, +0.155]	–	+0.186 → -0.011
HSA const-theta	Core CPI	+0.071 [+0.030, +0.110]	+0.031 [+0.016, +0.046]	988.0	+0.047 [-0.016, +0.110]	–	+0.177 → -0.015
HSA const-theta	PPI	+0.079 [-0.141, +0.299]	+0.010 [-0.028, +0.047]	1.1	-0.037 [-0.346, +0.283]	–	+0.111 → +0.053
HSA full	Headline CPI	+0.082 [+0.008, +0.153]	+0.034 [+0.006, +0.061]	14.1	+0.065 [-0.058, +0.198]	-0.016 [-0.054, +0.022]	+0.170 → +0.004
HSA full	Core CPI	+0.067 [+0.026, +0.110]	+0.029 [+0.012, +0.046]	99.9	+0.054 [-0.027, +0.136]	-0.009 [-0.041, +0.021]	+0.154 → -0.010
HSA full	PPI	+0.086 [-0.128, +0.303]	+0.010 [-0.029, +0.047]	1.1	+0.001 [-0.322, +0.330]	-0.005 [-0.045, +0.033]	+0.114 → +0.062

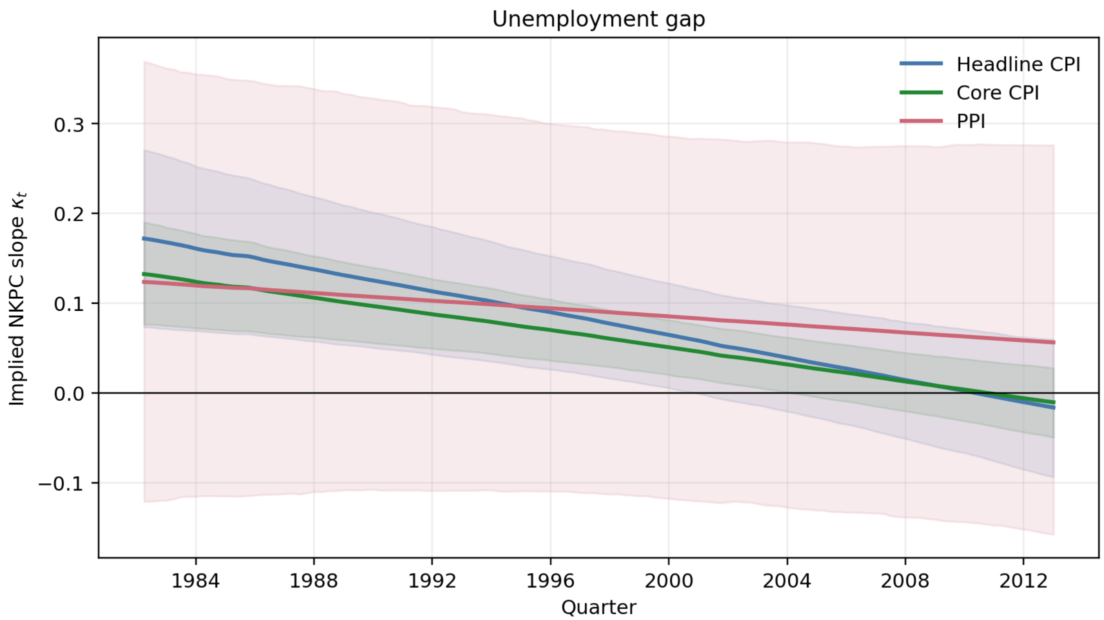

Figure 5: Interpolated design: time-varying slope κ_t for the negative unemployment gap. Compare Figure 2: the paths are close, but the bands here do not include the uncertainty from the 93 firm-count values that were interpolated rather than observed.

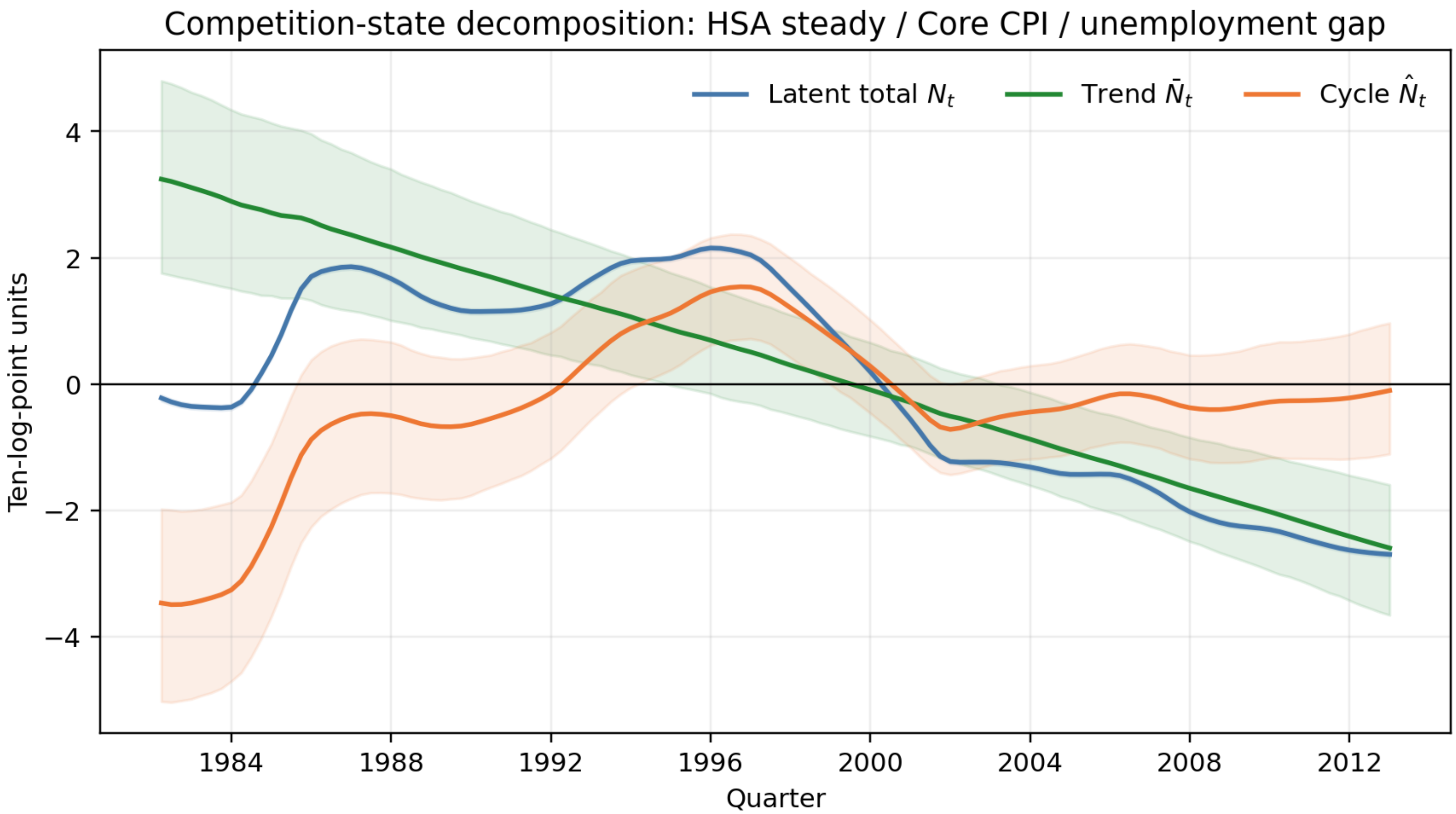

Figure 6: Interpolated design: firm-count state decomposition. The cycle $\hat{N}_t$ reaches ±3.4 ten-log-point units—a ±34 log-point "cyclical" swing in the number of listed firms—which is the level ridge, not an economic cycle. Compare Figure 3.

7.3 Prior sensitivity under interpolation

Table 12: Interpolated design: three-prior sensitivity for the negative unemployment gap. † marks cells outside the convergence criteria.

model	inflation	parameter	baseline	weak	tight
CES	Core CPI	κ	+0.039 [+0.009, +0.070]	+0.040 [+0.010, +0.070]	+0.038 [+0.003, +0.074]
CES	Headline CPI	κ	+0.054 [-0.009, +0.115]	+0.053 [-0.008, +0.114]	+0.059 [-0.003, +0.121]
CES	PPI	κ	+0.082 [-0.118, +0.288]	+0.073 [-0.158, +0.305]	+0.096 [-0.053, +0.248]
HSA const-theta	Core CPI	δ	+0.031 [+0.016, +0.046]	+0.039 [+0.022, +0.060]	+0.017 [+0.004, +0.030]
HSA const-theta	Headline CPI	δ	+0.035 [+0.009, +0.061]	+0.069 [+0.028, +0.137]	+0.014 [-0.003, +0.030]
HSA const-theta	PPI	δ	+0.010 [-0.028, +0.047]	+0.084 [-0.052, +0.214]†	+0.004 [-0.015, +0.023]
HSA dynamic	Core CPI	κ	+0.032 [-0.003, +0.067]	+0.035 [-0.002, +0.071]†	+0.028 [-0.010, +0.065]
HSA dynamic	Headline CPI	κ	+0.052 [-0.025, +0.144]†	+0.075 [-0.015, +0.162]†	+0.036 [-0.029, +0.102]
HSA dynamic	PPI	κ	+0.050 [-0.165, +0.269]†	+0.081 [-0.206, +0.377]†	+0.071 [-0.088, +0.226]
HSA full	Core CPI	δ	+0.029 [+0.012, +0.046]†	+0.062 [+0.011, +0.164]†	+0.016 [+0.003, +0.030]†
HSA full	Headline CPI	δ	+0.034 [+0.006, +0.061]†	+0.040 [-0.141, +0.172]†	+0.013 [-0.003, +0.030]†
HSA full	PPI	δ	+0.010 [-0.029, +0.047]†	+0.068 [-0.085, +0.201]†	+0.004 [-0.016, +0.022]†
HSA steady	Core CPI	δ	+0.025 [+0.012, +0.042]	+0.042 [+0.018, +0.115]†	+0.015 [+0.004, +0.025]
HSA steady	Headline CPI	δ	+0.031 [+0.010, +0.053]	+0.058 [+0.021, +0.130]†	+0.017 [+0.001, +0.032]
HSA steady	PPI	δ	+0.012 [-0.026, +0.049]†	+0.075 [-0.053, +0.198]†	+0.004 [-0.015, +0.023]†

Table 13: Interpolated design: model-by-model convergence tally.

model	sampler	runs	warnings	state warnings	joint warnings	max $\hat{R}$	min bulk ESS
CES	conjugate Gibbs	16	0	0	0	1.002	2644
HSA const-theta	joint FFBS	15	5	3	5	1.016	115
HSA dynamic	joint FFBS	16	8	8	9	1.033	83
HSA full	Particle Gibbs	15	15	15	15	1.821	3
HSA steady	joint FFBS	15	5	2	5	1.014	166

Table 14: Interpolated design: scalar / state-path / derived-path diagnostics by model. Compare Table 23: the worst-mixing scalar here is the trend drift n in nearly every cell, which is the level ridge in the parameter block.

model	inflation	sampler	scalar Rhat	scalar ESS	worst scalar	n Rhat	n ESS	state Rhat	state ESS	derived Rhat	derived ESS	joint
CES	Headline CPI	conjugate Gibbs	1.001	2785	lambda_ez	–	–	–	–	–	–	OK
CES	Core CPI	conjugate Gibbs	1.001	2836	sigma_e	–	–	–	–	–	–	OK
CES	PPI	conjugate Gibbs	1.001	2995	phi_1	–	–	–	–	–	–	OK
HSA steady	Headline CPI	joint FFBS	1.002	643	n	1.002	643	1.002	727	1.001	2627	OK
HSA steady	Core CPI	joint FFBS	1.002	662	n	1.002	662	1.002	779	1.000	3215	OK
HSA steady	PPI	joint FFBS	1.002	358	n	1.002	358	1.003	452	1.002	2867	watch
HSA dynamic	Headline CPI	joint FFBS	1.023	98	n	1.023	98	1.023	102	–	–	watch
HSA dynamic	Core CPI	joint FFBS	1.009	577	n	1.009	577	1.006	683	–	–	OK
HSA dynamic	PPI	joint FFBS	1.015	200	n	1.015	200	1.014	207	–	–	watch
HSA const-theta	Headline CPI	joint FFBS	1.004	633	n	1.004	633	1.003	735	1.000	1905	OK
HSA const-theta	Core CPI	joint FFBS	1.003	713	n	1.003	713	1.004	827	1.002	2164	OK
HSA const-theta	PPI	joint FFBS	1.003	416	n	1.003	416	1.004	471	1.000	1265	OK
HSA full	Headline CPI	Particle Gibbs	1.179	8	n	1.179	8	1.528	4	1.010	1081	watch
HSA full	Core CPI	Particle Gibbs	1.327	5	n	1.327	5	2.561	3	1.049	28	watch
HSA full	PPI	Particle Gibbs	1.380	5	n	1.380	5	1.577	4	1.080	17	watch

A prior–data conflict that is specific to interpolation. The AR(2) prior, $\rho_1 \sim \mathcal{N}(0.5, \sigma_\rho^2)$ and $\rho_2 \sim \mathcal{N}(-0.5, \sigma_\rho^2)$ truncated to the stationary region, encodes a cycle of about 5.2 quarters with a two-quarter half-life, and does so identically under all three prior sets—they differ only in width. Under the mixed-frequency design the posterior is compatible with it: the ρ_1 posterior sits between -1.5 and $+1.6$ prior standard deviations of the prior mean. Under interpolation it sits $+2.7$ to $+7.1$ prior standard deviations away. The prior is not obviously wrong; the interpolated series is simply not the object the prior describes. One consequence is that the tight-prior cells inherit a prior–data tug-of-war in the AR(2) block, which is where their mixing failures come from.

8 Scope condition and the limits of identification

For honesty we state where the main result does *not* hold and what it can and cannot claim.

- **Not identified for PPI.** Switching to PPI, closer to the theory's left-hand side, makes δ include zero for all activity measures. Despite the measurement limitation that no PPI expectation exists, we cannot claim that the CPI relationship holds generally at the price-setting stage.

- **HSA full does not converge; const-theta now does, in part.** Every HSA-full cell is outside the acceptance criteria even under Particle Gibbs, so γ—which only HSA full estimates—remains undemonstrated. HSA const-theta is a different case: its earlier failure was a blocking artifact of the alternating FFBS, and under the exact joint FFBS 10 of its 15 cells now pass the *whole-run* coefficient rule under the interpolated design, against 0 under mixed frequency, where the AR(2) block fails for the reason below. Its δ agrees in sign and magnitude with HSA steady in both. That is a computational fix, not new identification.

- **The joint posterior does not converge, though δ does.** Every HSA cell in the main design is outside the whole-run acceptance criteria. The failure is concentrated in the AR(2) block: (ρ_1, ρ_2) is the worst-mixing scalar in 54 of the 61 HSA cells, and the latent-state paths fail alongside it. What does *not* fail is the coefficient the conclusions rest on. Applied to δ *alone* rather than to all 18 scalars at once, the same $\hat{R} \leq 1.01$ / bulk ESS ≥ 400 thresholds are met in 8 of the 9 HSA-steady baseline cells at a median bulk ESS of 2,882, and δ and κ_0 fail in only 10 and 9 of 61 cells here (5 and 3 under interpolation). We therefore report δ and κ_t as results and treat the estimated cycle $\hat{N}_t$, its AR(2) coefficients, and anything that depends on them as diagnostic only. Readers who want the whole-run criterion satisfied should treat every number in this paper as provisional.

The latent-level ridge, and where it comes from. The firm-count equation $N_t^{\text{obs}} = \bar{N}_t + \hat{N}_t + \nu_t$ constrains the *sum* of the two states but not the split. Adding a constant c to $\bar{N}$ and subtracting it from $\hat{N}$ leaves that equation, the random-walk trend equation, and—through $\kappa_0 \to \kappa_0 - \delta c$—the inflation equation all exactly unchanged. The only thing that resists is the AR(2) cycle equation, which carries no intercept and therefore pulls $\hat{N}$ toward mean zero with force $1 - \rho_1 - \rho_2$; the initial-state prior $s_0 \sim \mathcal{N}(0, 10I_3)$ resists weakly as well.

Earlier revisions of this paper described the resulting ridge as a limitation of the data. It is better described as a consequence of the interpolated design. Under interpolation the AR(2) is pushed to a near-unit root and $1 - \rho_1 - \rho_2$ collapses to 0.014, so the anchor effectively disappears and the posterior correlation between the two states reaches -0.9996. Under the mixed-frequency design used here the same quantity is 1.698 and the correlation is $+0.1957$. Across all 122 estimated HSA cells the two quantities correlate at $+0.93$, which identifies the mechanism (Section 7).

Two things follow. First, κ_0—the slope at average competition—is identified only up to the location normalisation of $\bar{N}$, and under interpolation that normalisation comes substantially from the initial-state prior rather than from data: the posterior correlation between the $\bar{N}$ level and κ_0 is -0.52 there against -0.35 here. Statements about the *level* of the Phillips-curve slope should be read with that in mind. Second, δ is not affected, because a level shift in $\bar{N}$ is absorbed by κ_0 and leaves the interaction $x_t \bar{N}_t$ that identifies δ unchanged. That is why the headline result is stable across designs while the state decomposition is not.

Not a causal effect. $\bar{N}_t$ is a near-monotone low-frequency trend, so on this design δ cannot be separated from other things that trended over the same decades: the Volcker monetary-policy regime change, the anchoring of inflation expectations, changes in labor-market institutions and unionization, globalization and import competition, changes in the frequency of price adjustment, or a simple time trend / structural break. The most we can say is that *the low-frequency decline in the firm count and the decline in inflation's sensitivity to the negative unemployment gap occurred together, with a sign consistent with the HSA theory.* We therefore avoid "competition causes flattening" and "we identify the mechanism."

9 Robustness

We judge robustness on three completed points: (i) how the sign, interval, and magnitude change under alternative priors, (ii) whether the result survives switching the price index to PPI, and (iii) whether the same conclusion holds using only converged runs. The prior check re-estimates *every cell* under baseline, weak, and tight priors (Table 16); it is a genuine re-estimation of the current runs, not a re-weighting of a single posterior. What the three sets actually change is in Table 15, and two features of it matter for reading the sweep. First, the three sets differ in *dispersion only* — every coefficient prior keeps its centre across the three — but the widening is not uniform: it multiplies most coefficient standard deviations by 2.5 and δ, γ by 5. (An earlier revision also moved the centre of θ, θ_0 from 0.1 to 0 between sets, which mixed a location change into the sweep; all three now centre them at 0, matching δ and γ and taking no prior position on the sign the theory predicts.) Second, "tight" holds every variance prior at the same implied mean while concentrating it, whereas "weak" sets the inverse-gamma shape to $a = 1$, for which the prior mean does not exist. That last choice is why the weak cells, not the tight ones, are where the sampler mixes worst.

HSA steady. The headline-CPI and core-CPI δ keep a positive sign under weak and baseline priors, but not their interval: core CPI moves $+0.034[+0.014, +0.093]$ (weak), $+0.026[+0.012, +0.041]$ (baseline), $+0.010[-0.007, +0.026]$ (tight), so the tight prior pulls the interval across zero. The result is *not* prior-invariant, and the paper does not claim otherwise. PPI includes zero under all three priors.

Because the three prior sets move every prior at once, we ran a factorial experiment crossing δ's own prior with the AR(2) prior while holding all others at baseline, to see which channel drives the sweep. **That experiment is run on the interpolated design**, which is the setting in which the AR(2) prior is most in tension with the data (Section 7) and therefore has the most scope to move δ; a smaller AR(2) contribution under mixed frequency follows a fortiori, but we have not estimated it. The answer is δ's own prior: tightening it alone accounts for 86% of the headline-CPI shrinkage and 66% of the core-CPI shrinkage, against 13% and 41% for the AR(2) prior (with a -7% interaction in the core cell, the two channels partly offsetting). The tension between the AR(2) prior and the interpolated data is therefore not what is moving δ.

Mixing is a separate matter, and it is the *weak* corner that breaks, not the tight one. For the main cell the worst scalar has bulk ESS 92 under baseline priors and 101 under tight, but 3 under weak, with $\hat{R}$ rising to 1.818. The weak set puts the inverse-gamma variance priors at $a = 1$ (Table 15), which has no prior mean; the posterior locations of $\sigma_u, \sigma_\varepsilon$ barely move relative to baseline, so what the weak set buys is not a different answer but a much heavier tail for the sampler to explore.

Model variants. HSA const-theta reproduces the main result almost exactly under the added entry term: $\delta = +0.031[+0.006, +0.056]$ for headline CPI and $\delta = +0.027[+0.010, +0.045]$ for core CPI, against $+0.031[+0.007, +0.056]$ and $+0.026[+0.012, +0.041]$ under HSA steady. The agreement is close enough that the firm-count-trend channel is evidently not being carried by the omission of the entry term. Under the interpolated design the same comparison also holds, and there 10 of the 15 const-theta cells additionally satisfy the whole-run coefficient rule; under mixed frequency 0 do, for the AR(2) reason above. HSA full remains outside the criteria in every cell under either design, so γ is not identified here. HSA dynamic is short of ESS in most cells once the trend drift n and the variance block are included in the rule.

Period robustness. The state equation assumes consecutive quarters. We do not use the old treatment that deletes an interior span and compresses what precedes and follows. The current 77 runs are full-sample comparisons; until a separate contiguous subsample estimation of

Table 15: What the three prior sets change. Coefficient rows are $N(\text{mean}, \text{sd}^2)$; variance rows are $\text{IG}(a, b)$ with the implied prior mean $b/(a-1)$ in small type. Read from `configs/priors_*.yaml`, so this table cannot disagree with the priors the sampler loaded.

Parameter	baseline	weak	tight
α	$N(0.5, 0.2^2)$	$N(0.5, 0.5^2)$	$N(0.5, 0.1^2)$
κ, κ_0	$N(0.1, 0.2^2)$	$N(0.1, 0.5^2)$	$N(0.1, 0.1^2)$
δ	$N(0, 0.02^2)$	$N(0, 0.1^2)$	$N(0, 0.01^2)$
θ, θ_0	$N(0, 0.2^2)$	$N(0, 0.5^2)$	$N(0, 0.1^2)$
γ	$N(0, 0.02^2)$	$N(0, 0.1^2)$	$N(0, 0.01^2)$
ϕ_1	$N(0.7, 0.2^2)$	$N(0.7, 0.5^2)$	$N(0.7, 0.1^2)$
ρ_1	$N(0.5, 0.2^2)$	$N(0.5, 0.5^2)$	$N(0.5, 0.1^2)$
ρ_2	$N(-0.5, 0.2^2)$	$N(-0.5, 0.5^2)$	$N(-0.5, 0.1^2)$
n	$N(0, 0.1^2)$	$N(0, 0.25^2)$	$N(0, 0.05^2)$
σ_u^2	$\text{IG}(2, 0.02)$ (mean 0.02)	$\text{IG}(1, 0.02)$ (mean undefined)	$\text{IG}(4, 0.06)$ (mean 0.02)
σ_ε^2	$\text{IG}(2, 0.01)$ (mean 0.01)	$\text{IG}(1, 0.01)$ (mean undefined)	$\text{IG}(4, 0.03)$ (mean 0.01)
σ_N^2	$\text{IG}(2, 0.01)$ (mean 0.01)	$\text{IG}(1, 0.01)$ (mean undefined)	$\text{IG}(4, 0.03)$ (mean 0.01)
σ_η^2	$\text{IG}(2, 2)$ (mean 2)	$\text{IG}(1, 1)$ (mean undefined)	$\text{IG}(4, 4)$ (mean 1.33333)

Table 16: Three-prior sensitivity under the main mixed-frequency design (negative-unemployment-gap specification): δ under baseline, weak, and tight priors, by model and price index. Each cell is a separate re-estimation, not a re-weighting. The interpolated counterpart is Table 12.

model	inflation	parameter	baseline	weak	tight
CES	Core CPI	κ	+0.039 [+0.009, +0.070]	+0.040 [+0.010, +0.070]	+0.038 [+0.003, +0.074]
CES	Headline CPI	κ	+0.054 [-0.009, +0.115]	+0.053 [-0.008, +0.114]	+0.059 [-0.003, +0.121]
CES	PPI	κ	+0.082 [-0.118, +0.288]	+0.073 [-0.158, +0.305]	+0.096 [-0.053, +0.248]
HSA const-theta	Core CPI	δ	+0.027 [+0.010, +0.045]†	+0.038 [+0.019, +0.060]†	+0.009 [-0.008, +0.026]†
HSA const-theta	Headline CPI	δ	+0.031 [+0.006, +0.056]†	+0.057 [+0.017, +0.115]†	+0.011 [-0.008, +0.029]†
HSA const-theta	PPI	δ	+0.010 [-0.028, +0.048]†	+0.080 [-0.042, +0.198]†	+0.002 [-0.017, +0.022]†
HSA dynamic	Core CPI	κ	+0.038 [+0.007, +0.069]†	+0.032 [-0.008, +0.069]†	+0.038 [+0.002, +0.072]†
HSA dynamic	Headline CPI	κ	+0.054 [-0.010, +0.116]†	+0.044 [-0.048, +0.169]†	+0.059 [-0.004, +0.120]†
HSA dynamic	PPI	κ	+0.057 [-0.164, +0.274]†	+0.026 [-0.243, +0.274]†	+0.082 [-0.071, +0.236]†
HSA full	Core CPI	δ	+0.030 [+0.013, +0.048]†	-0.047 [-0.205, +0.055]†	+0.009 [-0.008, +0.024]†
HSA full	Headline CPI	δ	+0.030 [+0.002, +0.055]†	+0.049 [-0.007, +0.100]†	+0.010 [-0.008, +0.029]†
HSA full	PPI	δ	+0.009 [-0.028, +0.047]†	+0.074 [-0.049, +0.204]†	+0.002 [-0.016, +0.021]†
HSA steady	Core CPI	δ	+0.026 [+0.012, +0.041]†	+0.034 [+0.014, +0.093]†	+0.010 [-0.007, +0.026]†
HSA steady	Headline CPI	δ	+0.031 [+0.007, +0.056]†	+0.054 [+0.021, +0.111]†	+0.011 [-0.007, +0.029]†
HSA steady	PPI	δ	+0.010 [-0.027, +0.046]†	+0.079 [-0.038, +0.196]†	+0.002 [-0.017, +0.021]†

the corrected revision is done, we do not include pre-2008 or post-1988 numbers in the conclusions.

Error structure (pending). A fourth robustness question is whether the conclusion survives replacing the i.i.d. inflation disturbance with the invertible MA(3) implied by overlapping four-quarter inflation. The companion implementation has smoke-tested cells but not yet the full production run-set, so it is not evidence used in the conclusions or headline tables.

10 Conclusion

Over 1982–2012 in the United States, the low-frequency decline in the number of listed firms and the decline in inflation's sensitivity to the negative unemployment gap moved together. We read this as theory-consistent time-series evidence, not a causal effect.

- **Sign consistent with HSA.** For the core-CPI negative-unemployment-gap specification $\delta = +0.026[+0.012, +0.041]$ ($\mathrm{BF}_{10} = 116.7$); a falling firm count implies a flatter slope, as the theory predicts.
- **Economically sizable flattening.** The implied κ_t falls from $+0.126$ to -0.007; the start-to-end response to a one-point negative unemployment gap differs by about 0.13 point.
- **Core CPI aligns the output gap.** Under core CPI the HP output gap shows the same positive sign; BN stays inconclusive.
- **Not confirmed for PPI.** The negative-unemployment-gap specification gives $\delta = +0.010[-0.027, +0.046]$, with the interval including zero for all activity measures, so the relationship is not a general upstream (price-setting-stage) effect.
- **No CES-slope bias found.** The theory's predicted negative bias of the CES slope is not supported: both the CES–HSA difference and the HSA-implied omitted-variable bias have intervals including zero, with, if anything, positive point estimates.
- **Cyclical-entry channel unidentified.** The theory predicts a positive entry coefficient θ in $-\theta_t \hat{N}_t$. In the main design HSA const-theta gives $+0.004[-0.182, +0.191]$ for headline CPI ($\Pr(\theta > 0) = 0.51$) and $-0.015[-0.141, +0.107]$ for core CPI ($\Pr(\theta > 0) = 0.34$): both centred essentially on zero, with posteriors that sit on top of the prior in Figure 11. The data do not move θ. An earlier revision reported these two cells as carrying *opposite* signs; that contrast was largely an artefact of a prior centred at the theory's preferred $+0.1$, and it does not survive centring the prior at zero. What does survive is a dependence on choices that should not matter: under interpolation the same core-CPI cell gives $+0.047[-0.016, +0.110]$ with $\Pr(\theta > 0) = 0.93$, against $-0.015[-0.141, +0.107]$ here, and within the interpolated design the sign flips between activity measures ($+0.047[-0.016, +0.110]$ with the unemployment gap versus $-0.050[-0.097, +0.003]$ with the BN output gap, the latter putting 97% of its mass on the theory-wrong side). We claim no support for this channel. γ is estimated only in HSA full, which converges in no cell under either design.
- **The result does not depend on the interpolation, but the reported precision does.** Estimating with the annual firm count PCHIP-interpolated to quarterly and used as data gives $\delta = +0.025[+0.012, +0.042]$ against $+0.026[+0.012, +0.041]$ here—the same number to within a rounding step. What the interpolation changes is everything around it: it drives the firm-count measurement error from 0.069 to 0.023, forces the AR(2) cycle to a near-unit root, and through that opens a near-exact ridge between the trend and cycle states (posterior correlation -0.9996 against $+0.1957$). We report it in Section 7 as a comparison rather than as the main specification.
- **Identification caveat.** The variation is concentrated in the 1980s and $\bar{N}_t$ is nearly collinear with a linear trend, so δ is an association, not a causal effect.

11 Status and open items

Estimation status (auto-generated). The main results use the 77 runs of the mixed-frequency design, of which 61 fail the scalar acceptance rule (maximum $\hat{R} \leq 1.01$ and minimum bulk ESS ≥ 400 over every stored scalar). The interpolated comparison of Section 7 uses 77 runs, of which 33 fail the same rule. CES carries no latent firm-count state, so its 16 cells are shared across both designs. All headline estimates, tables, and figures are generated automatically from one complete saved run-set, revision `2026-08-theta-centred`. The sample-window correction intentionally advances the code revision before the long re-estimation is run; until that deferred run completes, the compiled numerical results remain the last complete run-set. A subsequent report build selects only the new revision and refreshes the macros, so revisions are never mixed within one table or claim. Section 8 states which failures bear on the conclusions and which do not.

Settled in the displayed complete run-set. The positive δ and flattening κ_t for the CPI negative-unemployment-gap specification of HSA steady, and their insensitivity to the firm-count observation design; the positive sign of the core-CPI HP specification and the non-result of BN; the non-results of the three PPI activity specifications; the prior sensitivity of δ and its decomposition into δ's own prior versus the AR(2) prior; the economic magnitude; the CES–HSA slope-bias diagnostic via the inverse markup; and the diagnosis that the latent-level ridge is a property of the interpolated design rather than of the data.

Open (empirical / econometric).

- **Causal identification.** The horse race against a linear time trend cannot distinguish the two. Before any claim beyond association, add quadratic trends, structural-break dummies, and placebo interactions.
- **Expectation mismatch.** Beyond core-matched expected inflation, connect a PPI forecast series (e.g. the Livingston Survey) so the expectation term matches the price index.
- **Genuinely out-of-sample evaluation.** Table 6 now reports prequential LPD_1, WAIC and PSIS-LOO next to the plug-in score, but all four are computed on the estimation sample. A rolling out-of-sample exercise—re-estimating on an expanding window and scoring only genuinely future quarters—remains open, and would be the only way to settle whether the time-varying slope forecasts better or merely fits better. The PPI and HSA-full cells additionally have maximum Pareto $\hat{k} > 0.7$, so their LOO values should be replaced by exact or K-fold cross-validation.
- **Period robustness.** Re-estimate the current revision using only contiguous quarters. The pre-2008/post-1988 numbers from earlier revisions are not used for the conclusions of this paper.
- **Bayes-factor robustness.** In addition to the two-sided SDDR, report the one-sided ($\delta > 0$) evaluation and the KDE bandwidth sensitivity.
- **Annual-Q4 state block.** The marginal chains of δ, κ_0 are good, but ρ_1, ρ_2 and some time points of $\bar{N}_t, \hat{N}_t$ have low ESS / high Rhat. Consider reparameterization, longer independent chains, and priors that relax the weak identification of the state equation.

Open (computational).

- **Conditional marginal likelihood — now reported (Section 6.1), with two things still open.** The implementation integrates the latent states out exactly by the Kalman filter, uses the estimating model's own initial state, takes the AR(2) ordinate from the sampler's own initial-lag conditional, normalizes the truncated (ρ_1, ρ_2) prior and conditional over the stationary triangle, and factorizes the ordinate with explicit reduced Gibbs runs. What still limits it: (i) HSA full is excluded by construction — its bilinear latent term is not linear-Gaussian, so no exact conditional likelihood or Gibbs ordinate over its blocks exists, and the implementation raises rather than returning the posterior-*mean* plug-in an earlier version reported; (ii) the Monte Carlo standard error, 0.16 on the main cell, is dominated by the firm-count block's reduced runs and would need longer runs, not a better estimator, to shrink. Only the mixed-frequency design and baseline priors have been run.

- **Weak convergence of full.** HSA full remains outside the criteria in every cell under Particle Gibbs. HSA const-theta's failure, by contrast, was a blocking problem and is resolved by the exact joint FFBS; that is a computational fix and is not evidence about identification.

- **Weak identification of γ.** An identification strategy is needed to relax the collinearity between $\hat{N}\bar{N}$ and $\hat{N}$.

- **Latent-level ridge.** $\bar{N}_t$ and $\hat{N}_t$ have posterior correlation -0.9996 under interpolation and $+0.1957$ under mixed frequency; only their sum is pinned by the firm-count equation. κ_0 is therefore identified partly through the initial-state prior, which is held fixed across the prior sweep. δ, which loads on the interaction $x_t\bar{N}_t$, is not affected by this ridge.

A Appendix: results by output gap (mixed-frequency, main design)

The body takes the negative unemployment gap as the main activity measure and separates the output-gap results here. Because HP and BN estimate potential output differently, we do not combine them into a single "output gap" coefficient but report the full baseline results by price index (3) and model (5) for each definition. The slope is the constant κ for CES/HSA dynamic and the slope at average competition κ_0 for the other HSA models. δ is the sensitivity of the slope to the firm-count trend, and entry denotes θ or θ_0.

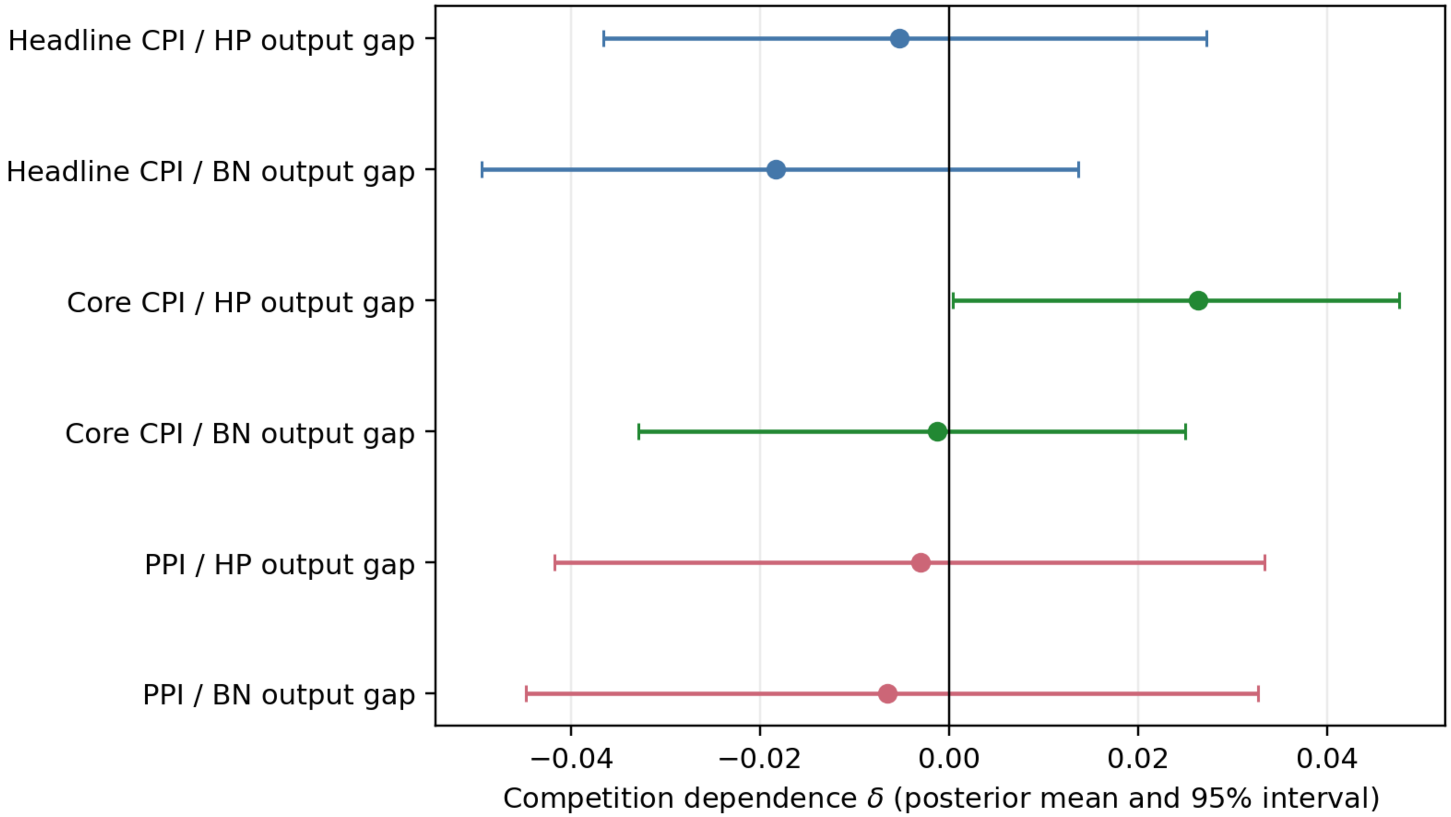

Figure 7: HSA-steady δ by output gap (posterior mean and 95% CI). HP and BN are separated and headline CPI, core CPI, and PPI are shown on the same scale.

A.1 HP output gap

For HSA steady the headline-CPI δ is near zero, while core CPI is positive:

$$\delta_{\text{Core,HP}} = +0.026[+0.000, +0.048], \qquad \text{BF}_{10} = 11.1.$$

The PPI $\delta = -0.003[-0.042, +0.033]$ includes zero. The positive competition dependence of the HP specification is therefore limited to core CPI and is not a result for price indices in general.

Table 17: HP output gap: baseline estimates by price index and model. † marks failure of the convergence criteria.

model	inflation	slope	δ	$\text{BF}_{10}(\delta)$	entry	γ	κ_t path
CES	Headline CPI	+0.139 [+0.045, +0.231]	–	–	–	–	–
CES	Core CPI	+0.091 [+0.044, +0.139]	–	–	–	–	–
CES	PPI	+0.213 [-0.053, +0.474]	–	–	–	–	–
HSA steady	Headline CPI	+0.139 [+0.047, +0.233]†	-0.005 [-0.037, +0.027]†	0.9	–	–	+0.134 → +0.153
HSA steady	Core CPI	+0.077 [+0.024, +0.128]†	+0.026 [+0.000, +0.048]†	11.1	–	–	+0.149 → +0.008
HSA steady	PPI	+0.209 [-0.051, +0.473]†	-0.003 [-0.042, +0.033]†	1.0	–	–	+0.205 → +0.217
HSA dynamic	Headline CPI	+0.140 [+0.049, +0.228]†	–	–	+0.013 [-0.222, +0.268]†	–	–
HSA dynamic	Core CPI	+0.090 [+0.044, +0.135]†	–	–	-0.027 [-0.219, +0.219]†	–	–
HSA dynamic	PPI	+0.158 [-0.116, +0.424]†	–	–	+0.036 [-0.316, +0.371]†	–	–
HSA const-theta	Headline CPI	+0.139 [+0.048, +0.230]†	-0.006 [-0.039, +0.029]†	0.9	+0.021 [-0.211, +0.272]†	–	+0.136 → +0.154
HSA const-theta	Core CPI	+0.074 [+0.024, +0.125]†	+0.027 [+0.007, +0.047]†	12.7	-0.032 [-0.118, +0.040]†	–	+0.148 → +0.003
HSA const-theta	PPI	+0.207 [-0.054, +0.467]†	-0.004 [-0.041, +0.035]†	1.0	+0.036 [-0.319, +0.400]†	–	+0.202 → +0.217
HSA full	Headline CPI	+0.136 [+0.047, +0.227]†	-0.007 [-0.042, +0.026]†	0.9	+0.023 [-0.275, +0.322]†	-0.000 [-0.039, +0.039]†	+0.134 → +0.155
HSA full	Core CPI	+0.073 [+0.011, +0.135]†	+0.017 [-0.021, +0.043]†	2.0	-0.007 [-0.240, +0.259]†	-0.005 [-0.042, +0.033]†	+0.121 → +0.030
HSA full	PPI	+0.211 [-0.056, +0.474]†	-0.004 [-0.042, +0.035]†	1.0	+0.028 [-0.317, +0.379]†	+0.000 [-0.037, +0.039]†	+0.206 → +0.221

A.2 BN output gap

For HSA steady the δ point estimates are negative for headline CPI, positive for core CPI, and negative for PPI, but all three 95% credible intervals include zero (headline $-0.018[-0.050, +0.014]$, core $-0.001[-0.033, +0.025]$, PPI $-0.006[-0.045, +0.033]$). Each BF_{10} is also about one, so the BN specification provides no evidence for a competition dependence of the slope.

Table 18: BN output gap: baseline estimates by price index and model. † marks failure of the convergence criteria.

model	inflation	slope	δ	$BF_{10}(\delta)$	entry	γ	κ_t path
CES	Headline CPI	+0.059 [-0.030, +0.146]	–	–	–	–	–
CES	Core CPI	+0.033 [-0.016, +0.081]	–	–	–	–	–
CES	PPI	+0.188 [-0.055, +0.432]	–	–	–	–	–
HSA steady	Headline CPI	+0.055 [-0.032, +0.138]†	-0.018 [-0.050, +0.014]†	1.6	–	–	+0.033 → +0.102
HSA steady	Core CPI	+0.029 [-0.022, +0.080]†	-0.001 [-0.033, +0.025]†	0.7	–	–	+0.038 → +0.033
HSA steady	PPI	+0.188 [-0.063, +0.436]†	-0.006 [-0.045, +0.033]†	1.1	–	–	+0.177 → +0.205
HSA dynamic	Headline CPI	+0.059 [-0.023, +0.142]†	–	–	+0.019 [-0.218, +0.258]†	–	–
HSA dynamic	Core CPI	+0.031 [-0.016, +0.079]†	–	–	-0.056 [-0.303, +0.239]†	–	–
HSA dynamic	PPI	+0.160 [-0.083, +0.403]†	–	–	+0.031 [-0.320, +0.384]†	–	–
HSA const-theta	Headline CPI	+0.054 [-0.034, +0.137]†	-0.019 [-0.052, +0.013]†	1.6	+0.019 [-0.223, +0.269]†	–	+0.035 → +0.103
HSA const-theta	Core CPI	+0.029 [-0.027, +0.084]†	-0.001 [-0.033, +0.026]†	0.8	-0.052 [-0.305, +0.232]†	–	+0.042 → +0.033
HSA const-theta	PPI	+0.185 [-0.055, +0.433]†	-0.007 [-0.044, +0.032]†	1.0	+0.033 [-0.311, +0.370]†	–	+0.176 → +0.204
HSA full	Headline CPI	+0.055 [-0.034, +0.143]†	-0.018 [-0.052, +0.015]†	1.5	+0.013 [-0.270, +0.285]†	-0.000 [-0.039, +0.040]†	+0.038 → +0.102
HSA full	Core CPI	+0.028 [-0.023, +0.078]†	-0.000 [-0.031, +0.026]†	0.8	-0.026 [-0.268, +0.288]†	-0.003 [-0.039, +0.034]†	+0.041 → +0.029
HSA full	PPI	+0.185 [-0.054, +0.430]†	-0.007 [-0.046, +0.030]†	1.0	+0.027 [-0.341, +0.376]†	+0.001 [-0.036, +0.039]†	+0.176 → +0.202

How to read the whole. HP core CPI agrees in direction with the CPI result for the negative unemployment gap, but HP headline, HP PPI, and all price indices of BN do not support it. And because HSA full has every cell, and const-theta most cells, outside the convergence criteria, we do not count sign agreement in the complex models as robustness evidence. The output-gap results are not a claim that "the result always holds across activity measures" but a sensitivity analysis showing which combinations of measurement and price index the result depends on.

B Appendix: results by output gap (interpolated design, for comparison)

We apply the same mixed-frequency estimation as Section 4 to the HP and BN output gaps. Each table gives the full baseline results by price index (3) and model (5); only the CES row is a common reference that does not depend on the firm-count observation scheme. We first compare the HSA-steady δ on the same scale, then show the full coefficients and diagnostics by model.

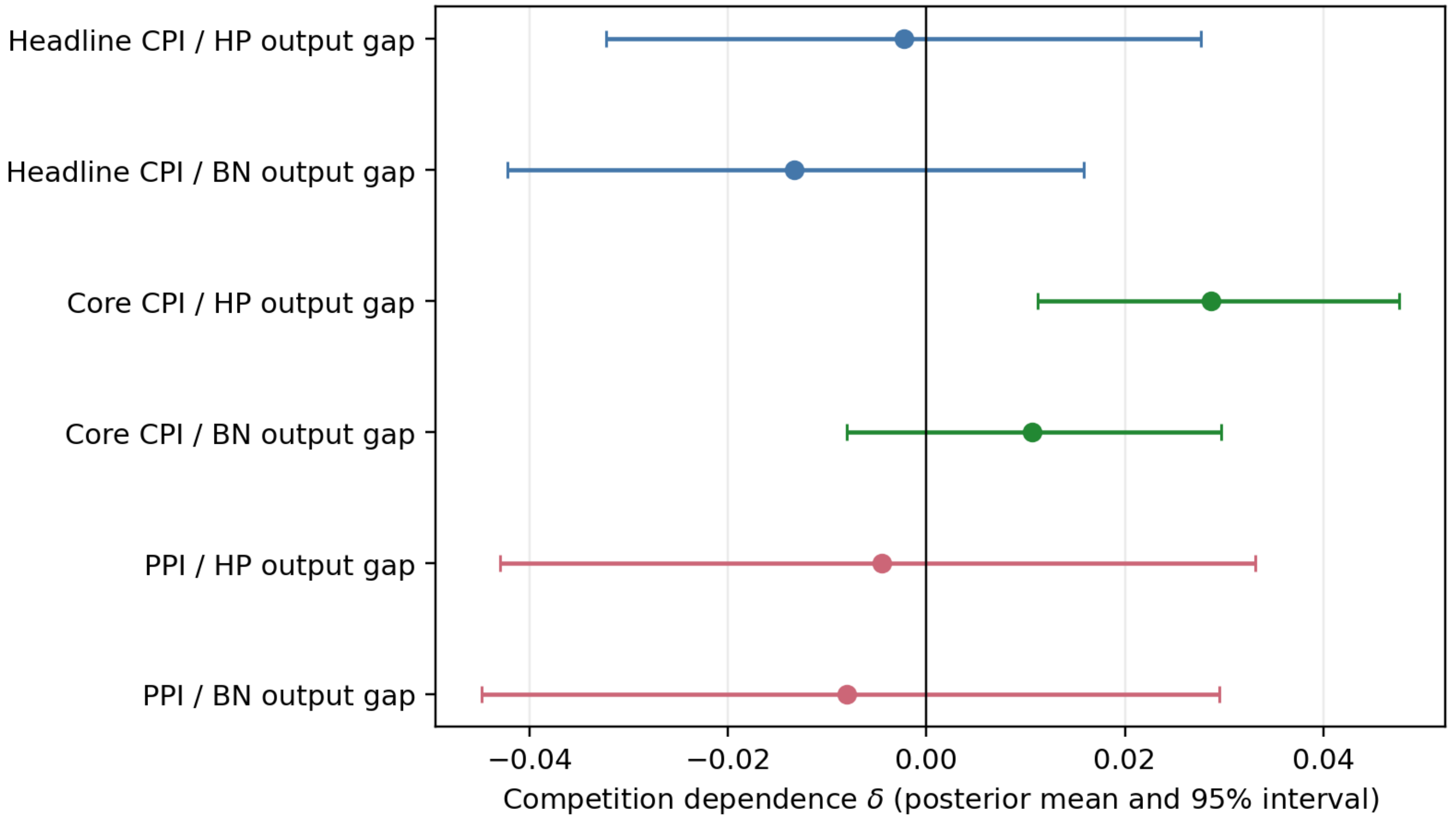

Figure 8: Interpolated design: HSA-steady δ by HP and BN output gap (posterior mean and 95% CI).

B.1 HP output gap

For the annual-Q4 HSA steady, headline CPI gives $\delta = -0.005[-0.037, +0.027]$ ($\mathrm{BF}_{10} = 0.9$), core CPI gives $\delta = +0.026[+0.000, +0.048]$ ($\mathrm{BF}_{10} = 11.1$), and PPI gives $\delta = -0.003[-0.042, +0.033]$ ($\mathrm{BF}_{10} = 1.0$).

Table 19: Interpolated design: HP output gap, baseline estimates by price index and model.

model	inflation	slope	δ	$BF_{10}(\delta)$	entry	γ	κ_t path
CES	Headline CPI	+0.139 [+0.045, +0.231]	–	–	–	–	–
CES	Core CPI	+0.091 [+0.044, +0.139]	–	–	–	–	–
CES	PPI	+0.213 [-0.053, +0.474]	–	–	–	–	–
HSA steady	Headline CPI	+0.143 [+0.046, +0.237]	-0.002 [-0.032, +0.028]	0.7	–	–	+0.136 → +0.148
HSA steady	Core CPI	+0.069 [+0.016, +0.120]	+0.029 [+0.011, +0.048]	66.7	–	–	+0.162 → -0.006
HSA steady	PPI	+0.212 [-0.066, +0.484]	-0.004 [-0.043, +0.033]	1.0	–	–	+0.198 → +0.224
HSA dynamic	Headline CPI	+0.131 [+0.040, +0.225]†	–	–	-0.026 [-0.129, +0.107]†	–	–
HSA dynamic	Core CPI	+0.083 [+0.038, +0.129]	–	–	-0.028 [-0.075, +0.024]	–	–
HSA dynamic	PPI	+0.152 [-0.133, +0.429]	–	–	-0.048 [-0.335, +0.249]	–	–
HSA const-theta	Headline CPI	+0.134 [+0.028, +0.231]†	-0.005 [-0.035, +0.026]†	0.8	-0.032 [-0.138, +0.101]†	–	+0.114 → +0.149
HSA const-theta	Core CPI	+0.068 [+0.011, +0.126]	+0.028 [+0.009, +0.048]	34.1	-0.007 [-0.056, +0.046]	–	+0.157 → -0.005
HSA const-theta	PPI	+0.203 [-0.067, +0.470]†	-0.005 [-0.043, +0.032]†	1.0	-0.038 [-0.321, +0.280]†	–	+0.185 → +0.218
HSA full	Headline CPI	+0.129 [+0.035, +0.220]†	-0.008 [-0.038, +0.022]†	0.9	-0.002 [-0.124, +0.114]†	-0.022 [-0.057, +0.013]†	+0.103 → +0.148
HSA full	Core CPI	+0.063 [+0.011, +0.113]†	+0.024 [+0.007, +0.043]†	16.8	+0.018 [-0.067, +0.096]†	-0.013 [-0.042, +0.015]†	+0.147 → -0.003
HSA full	PPI	+0.211 [-0.062, +0.481]†	-0.005 [-0.044, +0.033]†	1.0	-0.010 [-0.305, +0.293]†	-0.003 [-0.042, +0.036]†	+0.196 → +0.224

B.2 BN output gap

For the annual-Q4 HSA steady, headline CPI gives $\delta = -0.018[-0.050, +0.014]$ ($BF_{10} = 1.6$), core CPI gives $\delta = -0.001[-0.033, +0.025]$ ($BF_{10} = 0.7$), and PPI gives $\delta = -0.006[-0.045, +0.033]$ ($BF_{10} = 1.1$).

Table 20: Interpolated design: BN output gap, baseline estimates by price index and model.

model	inflation	slope	δ	$BF_{10}(\delta)$	entry	γ	κ_t path
CES	Headline CPI	+0.059 [-0.030, +0.146]	–	–	–	–	–
CES	Core CPI	+0.033 [-0.016, +0.081]	–	–	–	–	–
CES	PPI	+0.188 [-0.055, +0.432]	–	–	–	–	–
HSA steady	Headline CPI	+0.065 [-0.019, +0.146]	-0.013 [-0.042, +0.016]	1.1	–	–	+0.023 → +0.098
HSA steady	Core CPI	+0.030 [-0.021, +0.081]	+0.011 [-0.008, +0.030]	0.9	–	–	+0.063 → +0.004
HSA steady	PPI	+0.190 [-0.058, +0.429]	-0.008 [-0.045, +0.030]	1.1	–	–	+0.164 → +0.211
HSA dynamic	Headline CPI	+0.055 [-0.027, +0.139]†	–	–	-0.079 [-0.169, +0.078]†	–	–
HSA dynamic	Core CPI	+0.034 [-0.012, +0.080]	–	–	-0.042 [-0.089, +0.007]	–	–
HSA dynamic	PPI	+0.155 [-0.096, +0.399]†	–	–	-0.088 [-0.372, +0.239]†	–	–
HSA const-theta	Headline CPI	+0.062 [-0.027, +0.149]†	-0.012 [-0.037, +0.015]†	1.0	-0.080 [-0.176, +0.040]†	–	+0.019 → +0.095
HSA const-theta	Core CPI	+0.030 [-0.020, +0.080]	+0.015 [-0.002, +0.033]	1.9	-0.050 [-0.097, +0.003]	–	+0.083 → -0.007
HSA const-theta	PPI	+0.188 [-0.048, +0.417]†	-0.008 [-0.044, +0.029]†	1.1	-0.088 [-0.367, +0.219]†	–	+0.162 → +0.209
HSA full	Headline CPI	+0.071 [-0.016, +0.159]†	-0.015 [-0.042, +0.013]†	1.2	+0.006 [-0.131, +0.143]†	-0.031 [-0.068, +0.005]†	+0.018 → +0.108
HSA full	Core CPI	+0.045 [-0.003, +0.093]†	+0.012 [-0.005, +0.031]†	1.2	+0.019 [-0.060, +0.100]†	-0.033 [-0.063, -0.003]†	+0.083 → +0.013
HSA full	PPI	+0.188 [-0.046, +0.430]†	-0.010 [-0.047, +0.027]†	1.1	-0.101 [-0.358, +0.178]†	-0.009 [-0.047, +0.028]†	+0.152 → +0.214

These correspond one-to-one with Appendix A of the PCHIP version. In the comparison we read not only the point estimate but the 95% credible interval, the Bayes factor, and the convergence flag together. Where an interval widens under annual-Q4, that is the result of not treating the quarterly firm count as observed and propagating the interpolation uncertainty.

C Appendix: full model-by-model tables (negative unemployment gap)

This appendix collects the dense, auto-generated tables that support Sections 5 and 9: the full model-by-model coefficients for the negative unemployment gap and the whole-run convergence tally. (The three-prior sensitivity table now sits with the robustness discussion, Table 16.) The slope is κ for CES/HSA dynamic and κ_0 for the other models; entry is θ or θ_0; † marks cells outside the convergence criteria.

Table 21: Model-by-model results for the negative-unemployment-gap specification. Slope is κ for CES/HSA dynamic and κ_0 otherwise; entry is θ or θ_0. † marks cells that fail the convergence criteria and are not interpreted.

model	inflation	slope	δ	$BF_{10}(\delta)$	entry	γ	κ_t path
CES	Headline CPI	+0.054 [-0.009, +0.115]	–	–	–	–	–
CES	Core CPI	+0.039 [+0.009, +0.070]	–	–	–	–	–
CES	PPI	+0.082 [-0.118, +0.288]	–	–	–	–	–
HSA steady	Headline CPI	+0.078 [+0.012, +0.145]	+0.031 [+0.007, +0.056]	13.5	–	–	+0.141 → -0.004
HSA steady	Core CPI	+0.058 [+0.023, +0.092]	+0.026 [+0.012, +0.041]	116.7	–	–	+0.126 → -0.007
HSA steady	PPI	+0.087 [-0.119, +0.291]	+0.010 [-0.027, +0.046]	1.1	–	–	+0.104 → +0.063
HSA dynamic	Headline CPI	+0.054 [-0.010, +0.116]	–	–	+0.011 [-0.221, +0.230]	–	–
HSA dynamic	Core CPI	+0.038 [+0.007, +0.069]	–	–	-0.037 [-0.242, +0.214]	–	–
HSA dynamic	PPI	+0.057 [-0.164, +0.274]	–	–	+0.030 [-0.315, +0.371]	–	–
HSA const-theta	Headline CPI	+0.080 [+0.014, +0.145]	+0.031 [+0.006, +0.056]	11.5	+0.004 [-0.182, +0.191]	–	+0.135 → -0.002
HSA const-theta	Core CPI	+0.062 [+0.025, +0.105]	+0.027 [+0.010, +0.045]	40.2	-0.015 [-0.141, +0.107]	–	+0.134 → -0.007
HSA const-theta	PPI	+0.087 [-0.126, +0.300]	+0.010 [-0.028, +0.048]	1.1	+0.031 [-0.313, +0.377]	–	+0.102 → +0.062
HSA full	Headline CPI	+0.079 [+0.006, +0.151]	+0.030 [+0.002, +0.055]	7.6	+0.007 [-0.226, +0.233]	-0.002 [-0.042, +0.038]	+0.129 → +0.001
HSA full	Core CPI	+0.068 [+0.023, +0.113]	+0.030 [+0.013, +0.048]	119.8	+0.075 [-0.042, +0.180]	-0.009 [-0.041, +0.021]	+0.182 → -0.011
HSA full	PPI	+0.087 [-0.123, +0.290]	+0.009 [-0.028, +0.047]	1.1	+0.030 [-0.330, +0.387]	+0.000 [-0.039, +0.039]	+0.098 → +0.063

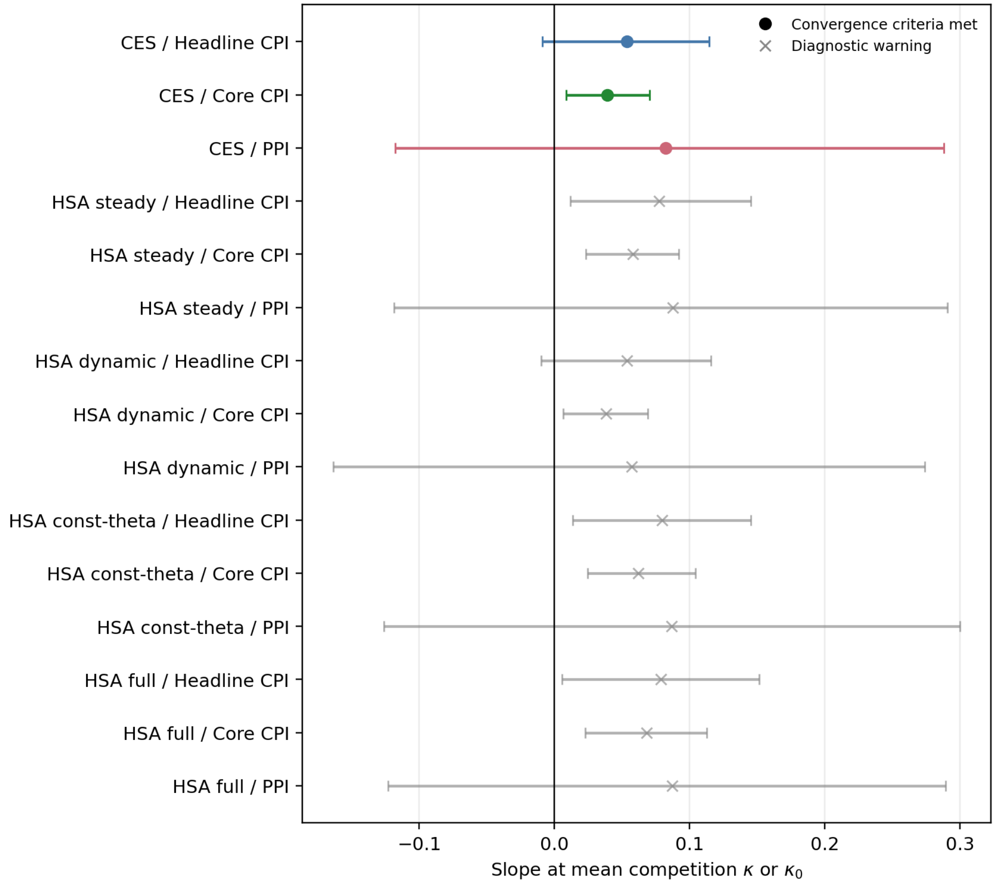

Figure 9: Slope at average competition for the negative-unemployment-gap specification. κ for CES/HSA dynamic and κ_0 otherwise. Circles are runs meeting the convergence criteria; gray crosses are runs to watch.

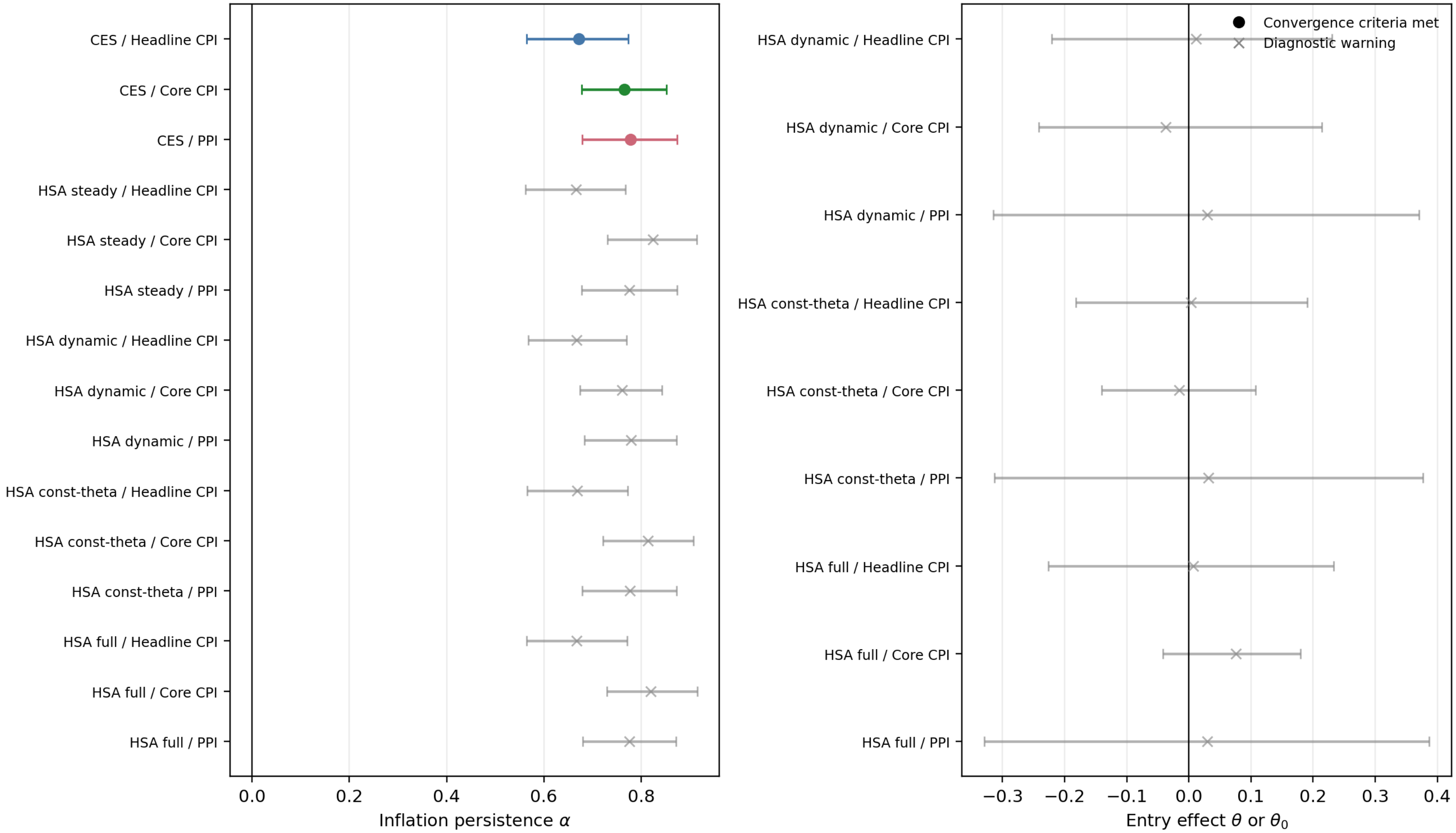

Figure 10: Inflation inertia α (left) and entry coefficient θ or θ_0 (right) for the negative-unemployment-gap specification. Circles and gray crosses are as in Figure 9.

Table 22: Model-by-model convergence tally across all 77 runs. "warnings" counts runs failing the *scalar* rule, "state warnings" the latent-path rule, and "joint warnings" at least one of the three groups (Section 3). The state-block column names the sampler that drew $\bar{N}_t, \hat{N}_t$.

model	sampler	runs	warnings	state warnings	joint warnings	max $\hat{R}$	min bulk ESS
CES	conjugate Gibbs	16	0	0	0	1.002	2644
HSA const-theta	joint FFBS	15	15	14	15	1.491	4
HSA dynamic	joint FFBS	16	16	15	16	1.397	5
HSA full	Particle Gibbs	15	15	15	15	1.831	3
HSA steady	joint FFBS	15	15	15	15	1.818	3

Table 23: Group-by-group convergence diagnostics for the negative-unemployment-gap baseline cells: scalar parameters, the trend drift n on its own, the latent-state paths $\bar{N}_t, \hat{N}_t$, and the derived paths κ_t, θ_t. "worst scalar" names the scalar with the lowest bulk ESS. Thresholds are $\hat{R} \leq 1.01$ and bulk ESS ≥ 400. Mixed-frequency (main) design. This table is the evidence behind the † and "OK (coef)" marks elsewhere: it shows whether a failure is driven by a coefficient of interest or by a nuisance term. Here the worst-mixing scalar is the AR(2) block in almost every cell, which for HSA steady is a nuisance object.

model	inflation	sampler	scalar Rhat	scalar ESS	worst scalar	n Rhat	n ESS	state Rhat	state ESS	derived Rhat	derived ESS	joint
CES	Headline CPI	conjugate Gibbs	1.001	2785	lambda_ez	–	–	–	–	–	–	OK
CES	Core CPI	conjugate Gibbs	1.001	2836	sigma_e	–	–	–	–	–	–	OK
CES	PPI	conjugate Gibbs	1.001	2995	phi_1	–	–	–	–	–	–	OK
HSA steady	Headline CPI	joint FFBS	1.032	56	rho_1	1.012	183	1.033	53	1.007	261	watch
HSA steady	Core CPI	joint FFBS	1.019	92	sigma_eps	1.006	685	1.013	108	1.007	195	watch
HSA steady	PPI	joint FFBS	1.059	27	rho_2	1.003	361	1.060	27	1.001	2946	watch
HSA dynamic	Headline CPI	joint FFBS	1.054	56	sigma_eps	1.010	409	1.070	49	–	–	watch
HSA dynamic	Core CPI	joint FFBS	1.038	50	rho_2	1.011	459	1.039	60	–	–	watch
HSA dynamic	PPI	joint FFBS	1.045	39	sigma_eps	1.010	195	1.044	38	–	–	watch
HSA const-theta	Headline CPI	joint FFBS	1.020	85	rho_1	1.002	359	1.016	75	1.002	378	watch
HSA const-theta	Core CPI	joint FFBS	1.491	4	rho_1	1.050	30	1.503	4	1.211	7	watch
HSA const-theta	PPI	joint FFBS	1.055	47	sigma_eps	1.011	240	1.052	49	1.001	3028	watch
HSA full	Headline CPI	Particle Gibbs	1.657	3	rho_2	1.038	38	1.675	3	1.050	28	watch
HSA full	Core CPI	Particle Gibbs	1.040	35	rho_1	1.005	235	1.021	72	1.006	353	watch
HSA full	PPI	Particle Gibbs	1.195	8	rho_2	1.022	99	1.116	13	1.002	2658	watch

D Appendix: every run behind this paper

The tables above report cells; this appendix reports *runs*. Table 24 and Table 25 list every posterior the paper reads — one row per (model, data spec, prior) cell, with the run identifier, the sampler that drew the state block, the estimation sample length, and the three convergence groups. Every number elsewhere in this paper is computed from exactly these runs and no others.

The run directory uses one stable path per (model, data spec, prior, observation design), named for that cell and carrying no timestamp. A superseded result can therefore remain on disk until that cell is re-estimated. The report builder admits only a complete set matching the requested estimation revision and sample signature; the tables below list exactly that admitted set. The revision string in each run's metadata records which vintage of the inputs produced it.

Table 24: Mixed-frequency (main) design: every run the paper reads. "state sampler" is the sampler that drew $\bar{N}_t, \hat{N}_t$; "max Rhat"/"min bulk ESS" are over the 18 scalars, and the state columns are over the latent paths.

model	data spec	prior	N frequency	T	run id	state sampler	max $\hat{R}$	min bulk ESS	state max $\hat{R}$	state min ESS
CES	inv_markup	baseline	PCHIP quarterly	124	20260806_005058_000	conjugate Gibbs	1.001065	2644.099394	–	–
CES	output_gap_bn	baseline	PCHIP quarterly	124	20260806_005058_001	conjugate Gibbs	1.000626	3066.241538	–	–
CES	output_gap_bn_core	baseline	PCHIP quarterly	124	20260806_005058_002	conjugate Gibbs	1.000873	2978.702688	–	–
CES	output_gap_bn_ppi	baseline	PCHIP quarterly	124	20260806_005058_003	conjugate Gibbs	1.000613	2964.867540	–	–
CES	output_gap_hp	baseline	PCHIP quarterly	124	20260806_005058_004	conjugate Gibbs	1.000014	2937.623117	–	–
CES	output_gap_hp_core	baseline	PCHIP quarterly	124	20260806_005058_005	conjugate Gibbs	1.000905	3110.610487	–	–
CES	output_gap_hp_ppi	baseline	PCHIP quarterly	124	20260806_005058_006	conjugate Gibbs	1.000652	3054.441076	–	–
CES	unemployment_gap	baseline	PCHIP quarterly	124	20260806_005058_007	conjugate Gibbs	1.000564	2784.663292	–	–
CES	unemployment_gap	tight	PCHIP quarterly	124	20260806_005058_008	conjugate Gibbs	1.001091	3113.785191	–	–
CES	unemployment_gap	weak	PCHIP quarterly	124	20260806_005058_009	conjugate Gibbs	1.000430	2905.817666	–	–
CES	unemployment_gap_core	baseline	PCHIP quarterly	124	20260806_005058_010	conjugate Gibbs	1.001098	2836.438142	–	–
CES	unemployment_gap_core	tight	PCHIP quarterly	124	20260806_005058_011	conjugate Gibbs	1.000575	2935.887420	–	–
CES	unemployment_gap_core	weak	PCHIP quarterly	124	20260806_005058_012	conjugate Gibbs	1.001859	3029.791368	–	–
CES	unemployment_gap_ppi	baseline	PCHIP quarterly	124	20260806_005058_013	conjugate Gibbs	1.000575	2994.818423	–	–
CES	unemployment_gap_ppi	tight	PCHIP quarterly	124	20260806_005058_014	conjugate Gibbs	1.000779	2932.791360	–	–
CES	unemployment_gap_ppi	weak	PCHIP quarterly	124	20260806_005058_015	conjugate Gibbs	1.000681	2923.966584	–	–
HSA const-theta	output_gap_bn	baseline	annual Q4	124	20260806_014651_000	joint FFBS	1.005436	82.677149	1.007307	96.905992
HSA const-theta	output_gap_bn_core	baseline	annual Q4	124	20260806_014651_001	joint FFBS	1.067604	32.261774	1.068758	30.413550
HSA const-theta	output_gap_bn_ppi	baseline	annual Q4	124	20260806_014651_002	joint FFBS	1.042891	63.260618	1.028762	72.396879
HSA const-theta	output_gap_hp	baseline	annual Q4	124	20260806_014651_003	joint FFBS	1.060912	62.745424	1.052115	73.108333
HSA const-theta	output_gap_hp_core	baseline	annual Q4	124	20260806_014651_004	joint FFBS	1.084409	26.055885	1.082357	21.589134
HSA const-theta	output_gap_hp_ppi	baseline	annual Q4	124	20260806_014651_005	joint FFBS	1.034656	100.156402	1.028116	94.108350
HSA const-theta	unemployment_gap	baseline	annual Q4	124	20260806_014651_006	joint FFBS	1.020228	84.729455	1.016400	74.706642
HSA const-theta	unemployment_gap	tight	annual Q4	124	20260806_014651_007	joint FFBS	1.007299	326.419501	1.008848	409.228755
HSA const-theta	unemployment_gap	weak	annual Q4	124	20260806_014651_008	joint FFBS	1.230852	7.099116	1.224682	7.448663
HSA const-theta	unemployment_gap_core	baseline	annual Q4	124	20260806_014651_009	joint FFBS	1.491121	3.810481	1.502852	3.798830
HSA const-theta	unemployment_gap_core	tight	annual Q4	124	20260806_014651_010	joint FFBS	1.010179	139.097819	1.015976	150.230034
HSA const-theta	unemployment_gap_core	weak	annual Q4	124	20260806_014651_011	joint FFBS	1.318620	5.078765	1.266621	5.570590
HSA const-theta	unemployment_gap_ppi	baseline	annual Q4	124	20260806_014651_012	joint FFBS	1.055183	46.616649	1.052170	49.203203
HSA const-theta	unemployment_gap_ppi	tight	annual Q4	124	20260806_014651_013	joint FFBS	1.005975	244.165383	1.005537	313.860669
HSA const-theta	unemployment_gap_ppi	weak	annual Q4	124	20260806_014651_014	joint FFBS	1.022423	52.340670	1.024278	64.495314
HSA dynamic	inv_markup	baseline	annual Q4	124	20260806_014651_015	joint FFBS	1.036999	43.200147	1.038964	51.887804
HSA dynamic	output_gap_bn	baseline	annual Q4	124	20260806_014651_016	joint FFBS	1.046325	44.632503	1.039497	52.213329
HSA dynamic	output_gap_bn_core	baseline	annual Q4	124	20260806_014651_017	joint FFBS	1.095236	17.283459	1.104814	14.607981
HSA dynamic	output_gap_bn_ppi	baseline	annual Q4	124	20260806_014651_018	joint FFBS	1.111584	15.392550	1.116466	14.275055
HSA dynamic	output_gap_hp	baseline	annual Q4	124	20260806_014651_019	joint FFBS	1.044367	57.390430	1.055299	56.514406
HSA dynamic	output_gap_hp_core	baseline	annual Q4	124	20260806_014651_020	joint FFBS	1.090983	22.729380	1.087372	23.556205
HSA dynamic	output_gap_hp_ppi	baseline	annual Q4	124	20260806_014651_021	joint FFBS	1.023222	73.843600	1.030252	64.538313
HSA dynamic	unemployment_gap	baseline	annual Q4	124	20260806_014651_022	joint FFBS	1.053600	56.457302	1.070332	49.045657
HSA dynamic	unemployment_gap	tight	annual Q4	124	20260806_014651_023	joint FFBS	1.004843	366.637422	1.005061	447.320212
HSA dynamic	unemployment_gap	weak	annual Q4	124	20260806_014651_024	joint FFBS	1.085960	27.781375	1.039513	41.600996
HSA dynamic	unemployment_gap_core	baseline	annual Q4	124	20260806_014651_025	joint FFBS	1.037855	50.225573	1.038961	59.741339
HSA dynamic	unemployment_gap_core	tight	annual Q4	124	20260806_014651_026	joint FFBS	1.005953	168.721261	1.005581	163.985309
HSA dynamic	unemployment_gap_core	weak	annual Q4	124	20260806_014651_027	joint FFBS	1.121457	11.887538	1.090909	15.607676
HSA dynamic	unemployment_gap_ppi	baseline	annual Q4	124	20260806_014651_028	joint FFBS	1.044775	38.557208	1.043619	37.992629
HSA dynamic	unemployment_gap_ppi	tight	annual Q4	124	20260806_014651_029	joint FFBS	1.010722	165.754515	1.014773	198.563847
HSA dynamic	unemployment_gap_ppi	weak	annual Q4	124	20260806_014651_030	joint FFBS	1.397423	4.531110	1.444511	4.108539
HSA full	output_gap_bn	baseline	annual Q4	124	20260806_014651_031	Particle Gibbs	1.390653	4.676377	1.345148	4.980986
HSA full	output_gap_bn_core	baseline	annual Q4	124	20260806_014651_032	Particle Gibbs	1.331312	4.930503	1.316121	5.259234
HSA full	output_gap_bn_ppi	baseline	annual Q4	124	20260806_014651_033	Particle Gibbs	1.129914	12.389113	1.107376	19.235112
HSA full	output_gap_hp	baseline	annual Q4	124	20260806_014651_034	Particle Gibbs	1.157404	9.131736	1.156177	10.922051
HSA full	output_gap_hp_core	baseline	annual Q4	124	20260806_014651_035	Particle Gibbs	1.125969	14.511910	1.144312	12.478141
HSA full	output_gap_hp_ppi	baseline	annual Q4	124	20260806_014651_036	Particle Gibbs	1.066389	37.191408	1.086690	25.290928
HSA full	unemployment_gap	baseline	annual Q4	124	20260806_014651_037	Particle Gibbs	1.656912	3.247835	1.675256	3.226696
HSA full	unemployment_gap	tight	annual Q4	124	20260806_014651_038	Particle Gibbs	1.041497	34.973684	1.065736	24.115655
HSA full	unemployment_gap	weak	annual Q4	124	20260806_014651_039	Particle Gibbs	1.508115	3.844695	1.649782	3.261590
HSA full	unemployment_gap_core	baseline	annual Q4	124	20260806_014651_040	Particle Gibbs	1.040340	34.646331	1.021083	71.796550
HSA full	unemployment_gap_core	tight	annual Q4	124	20260806_014651_041	Particle Gibbs	1.038744	40.522409	1.036685	51.196250
HSA full	unemployment_gap_core	weak	annual Q4	124	20260806_014651_042	Particle Gibbs	1.830589	2.907497	1.880853	2.907249
HSA full	unemployment_gap_ppi	baseline	annual Q4	124	20260806_014651_043	Particle Gibbs	1.194812	8.263471	1.115748	13.024073
HSA full	unemployment_gap_ppi	tight	annual Q4	124	20260806_014651_044	Particle Gibbs	1.127456	12.775618	1.083738	18.353386
HSA full	unemployment_gap_ppi	weak	annual Q4	124	20260806_014651_045	Particle Gibbs	1.386109	4.754040	1.731954	3.118685
HSA steady	output_gap_bn	baseline	annual Q4	124	20260806_014651_046	joint FFBS	1.059313	66.373180	1.053772	80.094504
HSA steady	output_gap_bn_core	baseline	annual Q4	124	20260806_014651_047	joint FFBS	1.043549	48.986816	1.039192	47.926737
HSA steady	output_gap_bn_ppi	baseline	annual Q4	124	20260806_014651_048	joint FFBS	1.046780	113.872027	1.053822	53.207877

Table 25: Interpolated (PCHIP) design: every run the paper reads. CES carries no latent firm-count state, so its cells are shared with the mixed-frequency design rather than re-estimated.

model	data spec	prior	N frequency	T	run id	state sampler	max $\hat{R}$	min bulk ESS	state max $\hat{R}$	state min ESS
CES	inv_markup	baseline	PCHIP quarterly	124	20260806_005058_000	conjugate Gibbs	1.001065	2644.099394	–	–
CES	output_gap_bn	baseline	PCHIP quarterly	124	20260806_005058_001	conjugate Gibbs	1.000626	3066.241538	–	–
CES	output_gap_bn_core	baseline	PCHIP quarterly	124	20260806_005058_002	conjugate Gibbs	1.000873	2978.702688	–	–
CES	output_gap_bn_ppi	baseline	PCHIP quarterly	124	20260806_005058_003	conjugate Gibbs	1.000613	2964.867540	–	–
CES	output_gap_hp	baseline	PCHIP quarterly	124	20260806_005058_004	conjugate Gibbs	1.000014	2937.623117	–	–
CES	output_gap_hp_core	baseline	PCHIP quarterly	124	20260806_005058_005	conjugate Gibbs	1.000905	3110.610487	–	–
CES	output_gap_hp_ppi	baseline	PCHIP quarterly	124	20260806_005058_006	conjugate Gibbs	1.000652	3054.441076	–	–
CES	unemployment_gap	baseline	PCHIP quarterly	124	20260806_005058_007	conjugate Gibbs	1.000564	2784.663292	–	–
CES	unemployment_gap	tight	PCHIP quarterly	124	20260806_005058_008	conjugate Gibbs	1.001091	3113.785191	–	–
CES	unemployment_gap	weak	PCHIP quarterly	124	20260806_005058_009	conjugate Gibbs	1.000430	2905.817666	–	–
CES	unemployment_gap_core	baseline	PCHIP quarterly	124	20260806_005058_010	conjugate Gibbs	1.001098	2836.438142	–	–
CES	unemployment_gap_core	tight	PCHIP quarterly	124	20260806_005058_011	conjugate Gibbs	1.000575	2935.887420	–	–
CES	unemployment_gap_core	weak	PCHIP quarterly	124	20260806_005058_012	conjugate Gibbs	1.001859	3029.791368	–	–
CES	unemployment_gap_ppi	baseline	PCHIP quarterly	124	20260806_005058_013	conjugate Gibbs	1.000575	2994.818423	–	–
CES	unemployment_gap_ppi	tight	PCHIP quarterly	124	20260806_005058_014	conjugate Gibbs	1.000779	2932.791360	–	–
CES	unemployment_gap_ppi	weak	PCHIP quarterly	124	20260806_005058_015	conjugate Gibbs	1.000681	2923.966584	–	–
HSA const-theta	output_gap_bn	baseline	PCHIP quarterly	124	20260806_005058_016	joint FFBS	1.008057	317.152816	1.007034	343.093312
HSA const-theta	output_gap_bn_core	baseline	PCHIP quarterly	124	20260806_005058_017	joint FFBS	1.004521	702.451416	1.004072	823.725928
HSA const-theta	output_gap_bn_ppi	baseline	PCHIP quarterly	124	20260806_005058_018	joint FFBS	1.004607	392.263249	1.004886	516.415335
HSA const-theta	output_gap_hp	baseline	PCHIP quarterly	124	20260806_005058_019	joint FFBS	1.016016	114.972399	1.014930	117.689446
HSA const-theta	output_gap_hp_core	baseline	PCHIP quarterly	124	20260806_005058_020	joint FFBS	1.006629	488.529288	1.005112	545.212828
HSA const-theta	output_gap_hp_ppi	baseline	PCHIP quarterly	124	20260806_005058_021	joint FFBS	1.016301	200.603988	1.013894	229.237188
HSA const-theta	unemployment_gap	baseline	PCHIP quarterly	124	20260806_005058_022	joint FFBS	1.003836	632.896199	1.002653	735.048266
HSA const-theta	unemployment_gap	tight	PCHIP quarterly	124	20260806_005058_023	joint FFBS	1.002557	1016.865118	1.001819	1816.581057
HSA const-theta	unemployment_gap	weak	PCHIP quarterly	124	20260806_005058_024	joint FFBS	1.001991	427.334301	1.004093	502.011397
HSA const-theta	unemployment_gap_core	baseline	PCHIP quarterly	124	20260806_005058_025	joint FFBS	1.002864	712.787113	1.003600	827.022717
HSA const-theta	unemployment_gap_core	tight	PCHIP quarterly	124	20260806_005058_026	joint FFBS	1.001425	1269.604790	1.002790	2006.925121
HSA const-theta	unemployment_gap_core	weak	PCHIP quarterly	124	20260806_005058_027	joint FFBS	1.001898	529.840824	1.002005	630.286592
HSA const-theta	unemployment_gap_ppi	baseline	PCHIP quarterly	124	20260806_005058_028	joint FFBS	1.002599	416.356945	1.004040	470.817317
HSA const-theta	unemployment_gap_ppi	tight	PCHIP quarterly	124	20260806_005058_029	joint FFBS	1.002484	1102.139003	1.001833	1939.840881
HSA const-theta	unemployment_gap_ppi	weak	PCHIP quarterly	124	20260806_005058_030	joint FFBS	1.005731	365.700103	1.005077	430.590804
HSA dynamic	inv_markup	baseline	PCHIP quarterly	124	20260806_005058_031	joint FFBS	1.007436	445.877291	1.006644	399.973570
HSA dynamic	output_gap_bn	baseline	PCHIP quarterly	124	20260806_005058_032	joint FFBS	1.013261	155.212511	1.011607	157.128501
HSA dynamic	output_gap_bn_core	baseline	PCHIP quarterly	124	20260806_005058_033	joint FFBS	1.001549	796.935392	1.002249	894.964042
HSA dynamic	output_gap_bn_ppi	baseline	PCHIP quarterly	124	20260806_005058_034	joint FFBS	1.012845	114.245153	1.013726	119.933432
HSA dynamic	output_gap_hp	baseline	PCHIP quarterly	124	20260806_005058_035	joint FFBS	1.033120	201.809273	1.036548	209.184636
HSA dynamic	output_gap_hp_core	baseline	PCHIP quarterly	124	20260806_005058_036	joint FFBS	1.004289	731.566916	1.002979	848.119650
HSA dynamic	output_gap_hp_ppi	baseline	PCHIP quarterly	124	20260806_005058_037	joint FFBS	1.004574	453.376350	1.006750	482.057377
HSA dynamic	unemployment_gap	baseline	PCHIP quarterly	124	20260806_005058_038	joint FFBS	1.022562	97.973516	1.023149	102.231415
HSA dynamic	unemployment_gap	tight	PCHIP quarterly	124	20260806_005058_039	joint FFBS	1.001134	1186.661864	1.001093	1810.497533
HSA dynamic	unemployment_gap	weak	PCHIP quarterly	124	20260806_005058_040	joint FFBS	1.018704	82.973886	1.016529	85.288066
HSA dynamic	unemployment_gap_core	baseline	PCHIP quarterly	124	20260806_005058_041	joint FFBS	1.009366	576.997725	1.005950	683.350036
HSA dynamic	unemployment_gap_core	tight	PCHIP quarterly	124	20260806_005058_042	joint FFBS	1.000817	1306.797549	1.004341	2079.955835
HSA dynamic	unemployment_gap_core	weak	PCHIP quarterly	124	20260806_005058_043	joint FFBS	1.001581	348.731155	1.002533	414.692081
HSA dynamic	unemployment_gap_ppi	baseline	PCHIP quarterly	124	20260806_005058_044	joint FFBS	1.015054	199.879472	1.013910	206.606585
HSA dynamic	unemployment_gap_ppi	tight	PCHIP quarterly	124	20260806_005058_045	joint FFBS	1.000989	1383.081649	1.001344	1965.192900
HSA dynamic	unemployment_gap_ppi	weak	PCHIP quarterly	124	20260806_005058_046	joint FFBS	1.029757	112.338257	1.031660	126.876797
HSA full	output_gap_bn	baseline	PCHIP quarterly	124	20260806_005058_047	Particle Gibbs	1.109659	11.687882	1.395240	4.325422
HSA full	output_gap_bn_core	baseline	PCHIP quarterly	124	20260806_005058_048	Particle Gibbs	1.702926	3.108883	3.292436	2.239110
HSA full	output_gap_bn_ppi	baseline	PCHIP quarterly	124	20260806_005058_049	Particle Gibbs	1.465609	3.830935	1.912930	2.779244
HSA full	output_gap_hp	baseline	PCHIP quarterly	124	20260806_005058_050	Particle Gibbs	1.029838	62.204722	1.342741	19.220888
HSA full	output_gap_hp_core	baseline	PCHIP quarterly	124	20260806_005058_051	Particle Gibbs	1.190211	7.233391	2.591325	3.540199
HSA full	output_gap_hp_ppi	baseline	PCHIP quarterly	124	20260806_005058_052	Particle Gibbs	1.496516	3.694653	1.997518	2.712734
HSA full	unemployment_gap	baseline	PCHIP quarterly	124	20260806_005058_053	Particle Gibbs	1.179220	8.110577	1.528431	3.862790
HSA full	unemployment_gap	tight	PCHIP quarterly	124	20260806_005058_054	Particle Gibbs	1.118134	11.620284	1.232851	7.089891
HSA full	unemployment_gap	weak	PCHIP quarterly	124	20260806_005058_055	Particle Gibbs	1.820603	2.921368	2.259530	2.532640
HSA full	unemployment_gap_core	baseline	PCHIP quarterly	124	20260806_005058_056	Particle Gibbs	1.326946	4.853296	2.560711	3.156237
HSA full	unemployment_gap_core	tight	PCHIP quarterly	124	20260806_005058_057	Particle Gibbs	1.057735	21.047337	1.149760	13.766661
HSA full	unemployment_gap_core	weak	PCHIP quarterly	124	20260806_005058_058	Particle Gibbs	1.802380	2.944714	2.043724	2.692925
HSA full	unemployment_gap_ppi	baseline	PCHIP quarterly	124	20260806_005058_059	Particle Gibbs	1.379859	4.535458	1.577264	3.524928
HSA full	unemployment_gap_ppi	tight	PCHIP quarterly	124	20260806_005058_060	Particle Gibbs	1.017051	73.032164	1.046697	33.147777
HSA full	unemployment_gap_ppi	weak	PCHIP quarterly	124	20260806_005058_061	Particle Gibbs	1.433339	4.029280	1.928601	2.809725
HSA steady	output_gap_bn	baseline	PCHIP quarterly	124	20260806_005058_062	joint FFBS	1.002322	660.814276	1.001436	812.504146
HSA steady	output_gap_bn_core	baseline	PCHIP quarterly	124	20260806_005058_063	joint FFBS	1.004897	548.758296	1.003971	779.990382
HSA steady	output_gap_bn_ppi	baseline	PCHIP quarterly	124	20260806_005058_064	joint FFBS	1.001558	554.844040	1.003980	637.700460

E Appendix: prior against posterior, every model and coefficient

The tables report intervals; these figures report the whole marginal posterior, and put it next to the prior it was estimated under. Each figure covers one activity measure under the main mixed-frequency design: models down the rows, the four coefficients across the columns, and the three price indices as the three curves in each panel. A dash marks a coefficient the model restricts away. The dashed line is the prior, read from each run's own saved prior block rather than from the config file, so a curve and its prior are always the pair that actually produced the draws.

Three things are worth reading off them. First, down the δ column, HSA steady, HSA const-theta and HSA full give an almost identical posterior: adding the entry term does not move the coefficient the conclusions rest on. Second, across every panel, the core-CPI curve is the sharpest and the PPI curve the widest — the same ordering the intervals show, but here it is visibly a difference in how much the data say rather than a difference in point estimates. Third, and most directly, the γ column sits on top of its own prior in all three price indices. The posterior is the prior; the data are silent about γ. That is the same conclusion Section 8 states from the convergence diagnostics, arrived at without reference to mixing.

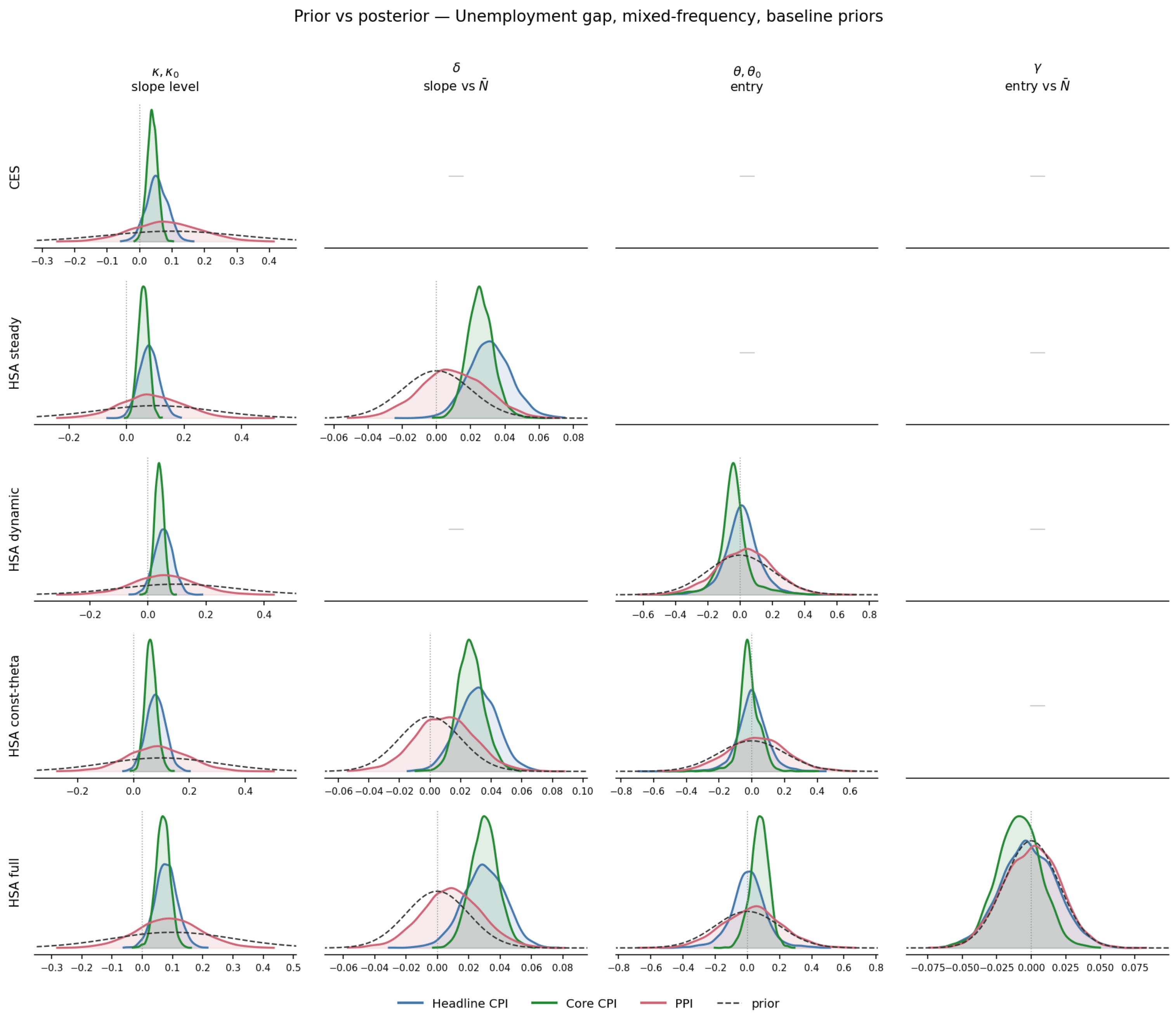

Figure 11: Prior against posterior, negative unemployment gap (the main activity measure), mixed-frequency design, baseline priors.

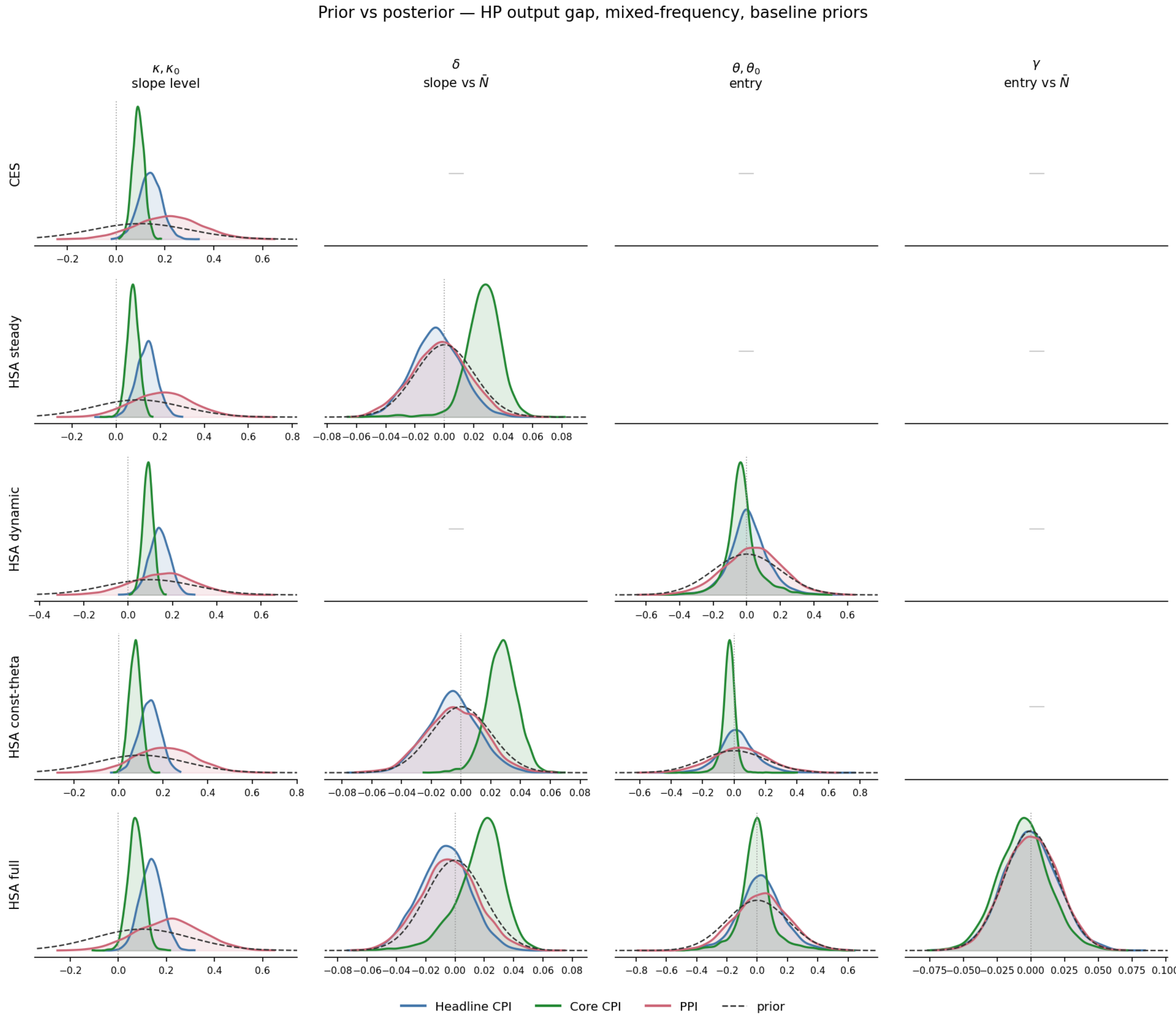

Figure 12: Prior against posterior, HP output gap, mixed-frequency design, baseline priors.

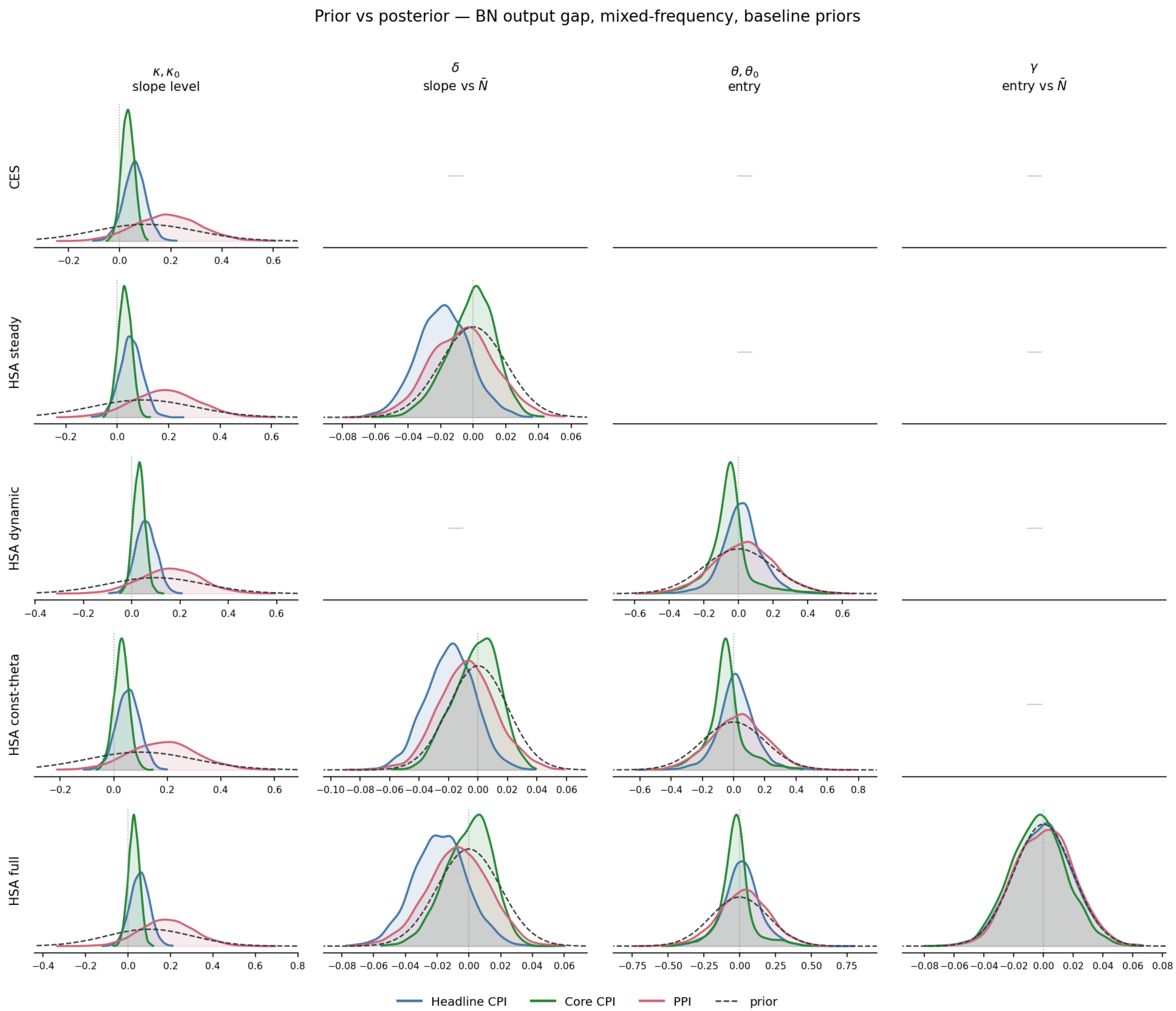

Figure 13: Prior against posterior, BN output gap, mixed-frequency design, baseline priors. The interpolated-design counterparts of all three figures are written to `results/figures/quarterly_interpolated/` by the same builder.